

NE 591: Special Topics In Nuclear Engineering I

Multi-Physics of Nuclear Reactors

UNIT 5: Novel Multi-Physics Modeling

Lecture 19: MULTI-SCALE SINGLE PHYSICS MODELING USING HI2LO INFORMATION SCHEMES AND IMPLEMENTATION WITHIN MULTI-PHYSICS CALCULATIONS

Hi2Lo BACKGROUND

- Multiple models of a physical process available to describe the same system each with different levels of fidelity
 - Presence of reduced order physics in lower fidelity models
 - Difference levels of accuracy due to coarser or finer grids
- Higher fidelity models better represent the physical process than lower fidelity ones, but are computationally expensive
- Idea of high-to-low (Hi2Lo) approach: combining relatively cheap lower fidelity model calculations with more costly high-fidelity simulations, to emulate the high-fidelity model prediction accuracy
 - Employ simulated data generated by higher-fidelity codes to inform lower-fidelity codes
 - Use simulator outputs with physical observations to build a predictive model and estimate unknown parameters that govern the behavior of the computer model

Hi2Lo BACKGROUND

- Designing such Hi2Lo approaches for each physics domain is not trivial because every domain will have its unique challenges, requiring different considerations for the Hi2Lo approaches.
- Two general paradigms for the development of Hi2Lo approaches can be identified:
 1. The offline approaches execute in advance numerous HiFi calculations to construct the Hi2Lo model that is then plugged into the multi-physics coupling.
 2. The online approaches iterate between the HiFi and LoFi models during the multi-physics coupling.
- The goal is to minimize the discrepancy between the high and low fidelity models for a specific domain of validity. We can formulate the problem in two different ways:
 - f_m^{HF} : high fidelity model
 - d_m : common parameters across fidelities
 - d_m^{HF} : high fidelity model specific parameters
 - f_m^{LF} : low fidelity model
 - d_m^{LF} : low fidelity model specific parameters
 - d_m^{HL} : low fidelity model specific parameters that can be predicted by the high-fidelity model
 - ϵ, ϵ_{HL} : Hi2Lo discrepancy

$$f_m^{HF}(d_m, d_m^{HF}) = f_m^{LF}(d_m, d_m^{LF}) + \epsilon(d_m, d_m^{HF}, d_m^{LF})$$

$$f_m^{HF}(d_m, d_m^{HF}) = f_m^{LF}(d_m, d_m^{LF}, d_m^{HL}) + \epsilon_{HL}(d_m, d_m^{HF}, d_m^{LF}) \quad 3$$

Hi2Lo BACKGROUND

- We use the second formulation to define the Hi2Lo approaches as methods that find an efficient representation g_m^{HL} of d_m^{HL} using the high-fidelity calculations

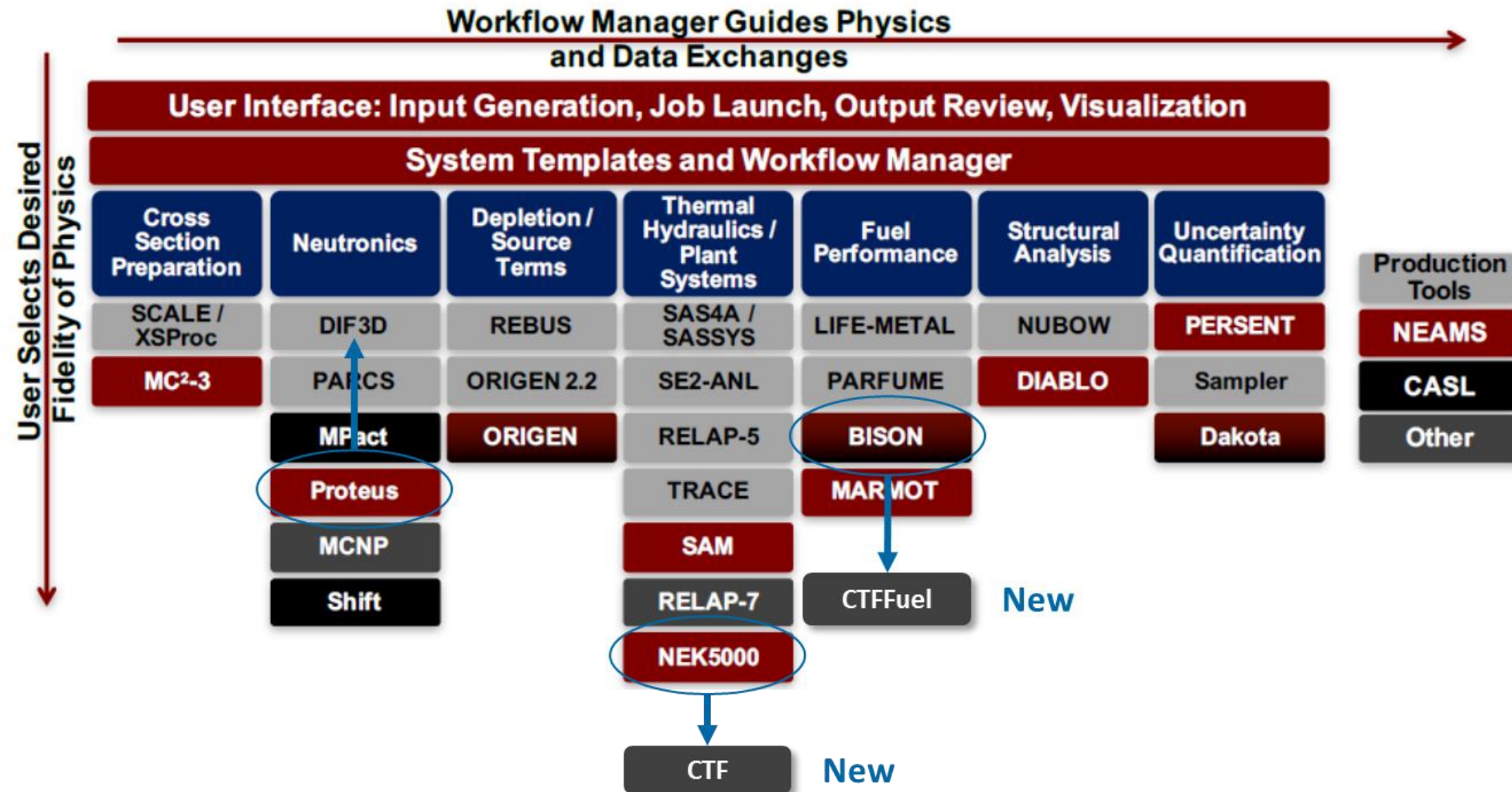
$$d_m^{HL} = g_m^{HL}(f_m^{HF}(d_m, d_m^{HF}))$$

$$f_m^{HF}(d_m, d_m^{HF}) = f_m^{LF}(d_m, d_m^{LF}, d_m^{HL}) + \varepsilon_{HL}(d_m, d_m^{HF}, d_m^{LF})$$

- This approach was used to develop a Hi2Lo approach for each physics in a general multi-physics framework:
 - Reactor physics $f_{rp}^{HF} \rightarrow f_{rp}^{LF}$
 - Thermal-hydraulics $f_{th}^{HF} \rightarrow f_{th}^{LF}$
 - Fuel performance $f_{fp}^{HF} \rightarrow f_{fp}^{LF}$
- The Hi2Lo multi-physics framework is used then for uncertainty analysis.

Hi2Lo BACKGROUND

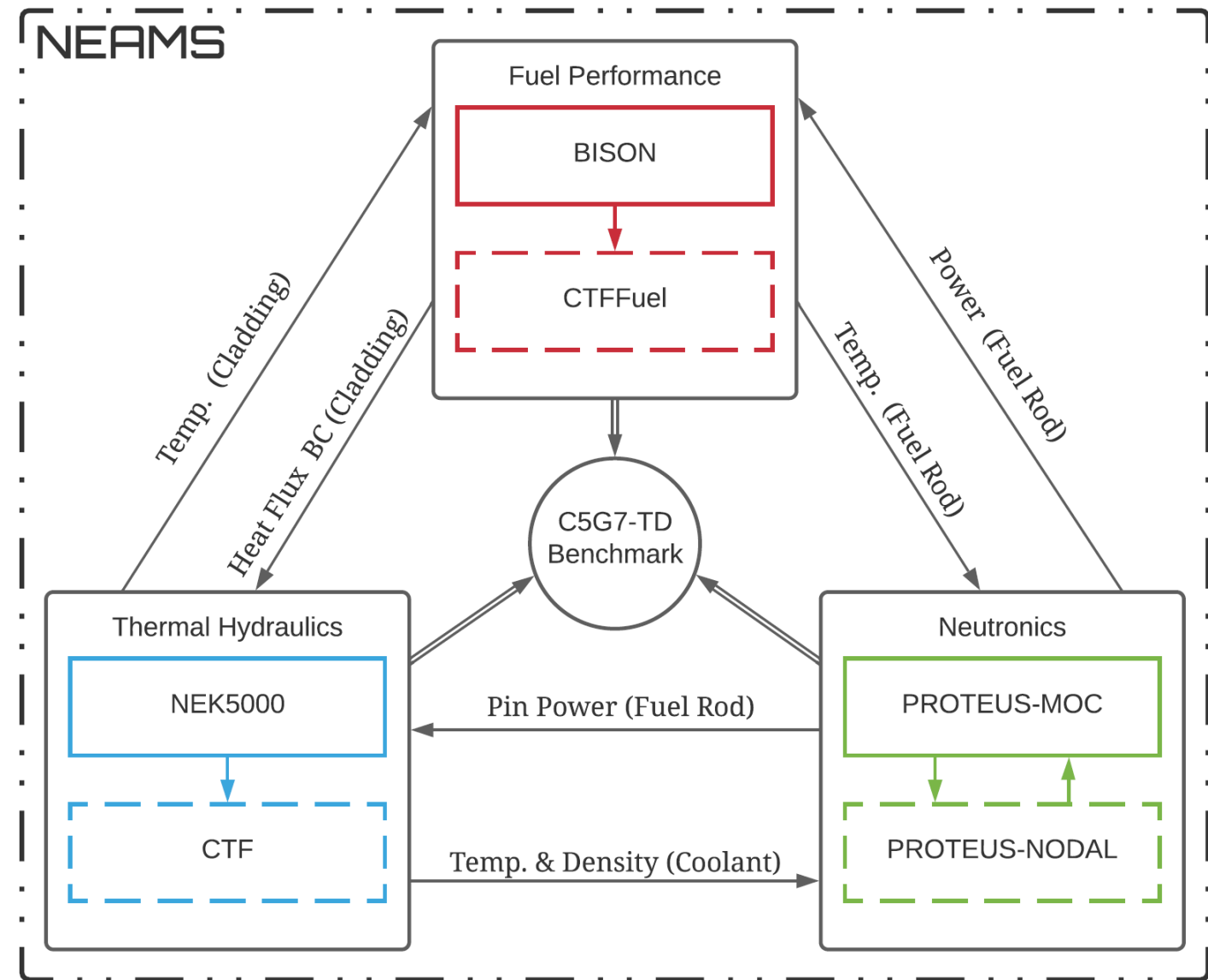
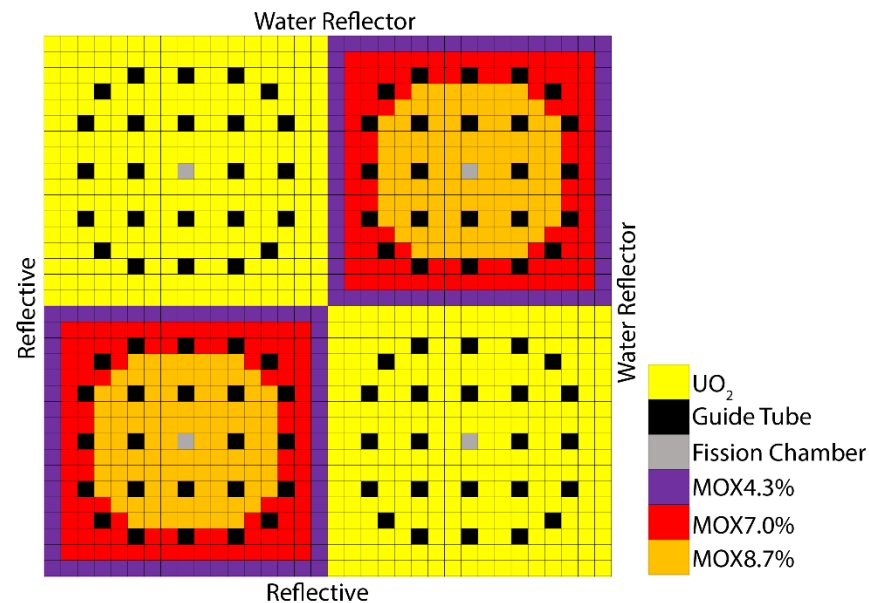
- NEAMS Workbench is intended to facilitate the transition from high-fidelity to conventional tools by providing a common user interface for model creation, review, execution, and visualization for integrated codes¹



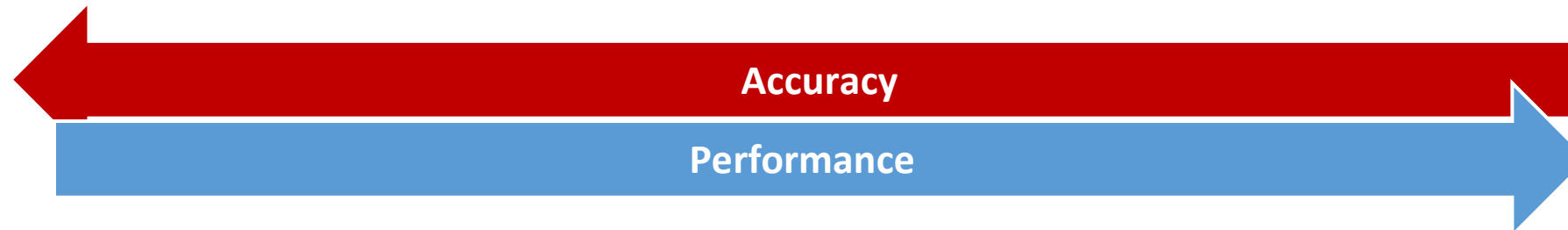
¹ B. Rearden, et al, 2017.

Hi2Lo BACKGROUND

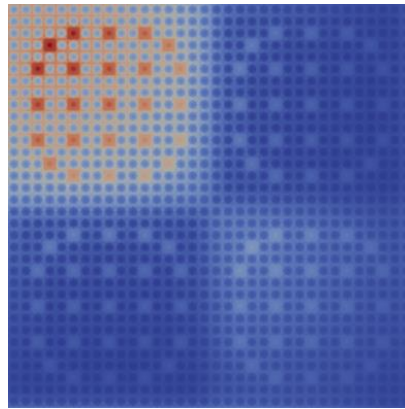
- The goal of Hi2Lo is to develop efficient high fidelity informed lower fidelity simulations that have reasonable computational cost so that they can be applied to:
 - Uncertainty analysis
 - Sensitivity analysis
 - Core design optimization
 - Core operation optimization
- Demonstration on the C5G7 benchmark core.



REACTOR PHYSICS Hi2Lo

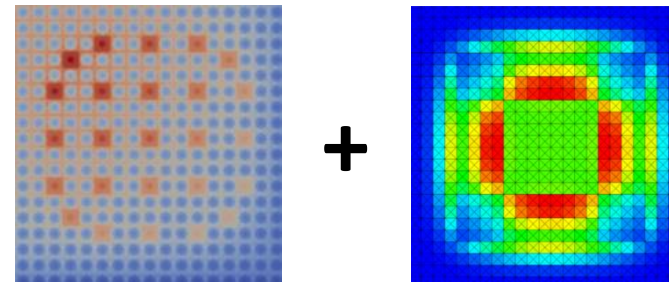


Transport



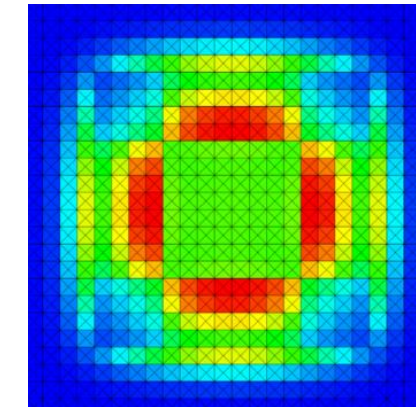
- Finite-element with SN
- Method of Characteristic
- Accelerated by CMFD, DSA ,...
- HiFi result

Hi2Lo



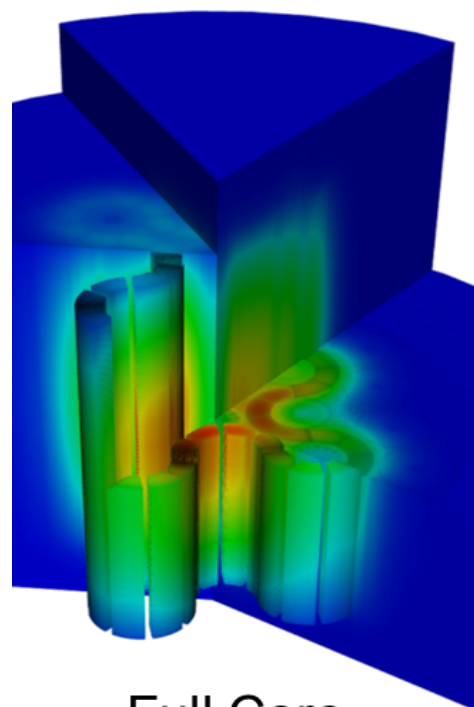
- LoFi-like performance & HiFi-like accuracy
- Iterate between LoFi & HiFi

Diffusion



- Homogenized geometry
- Diffusion method
- LoFi result

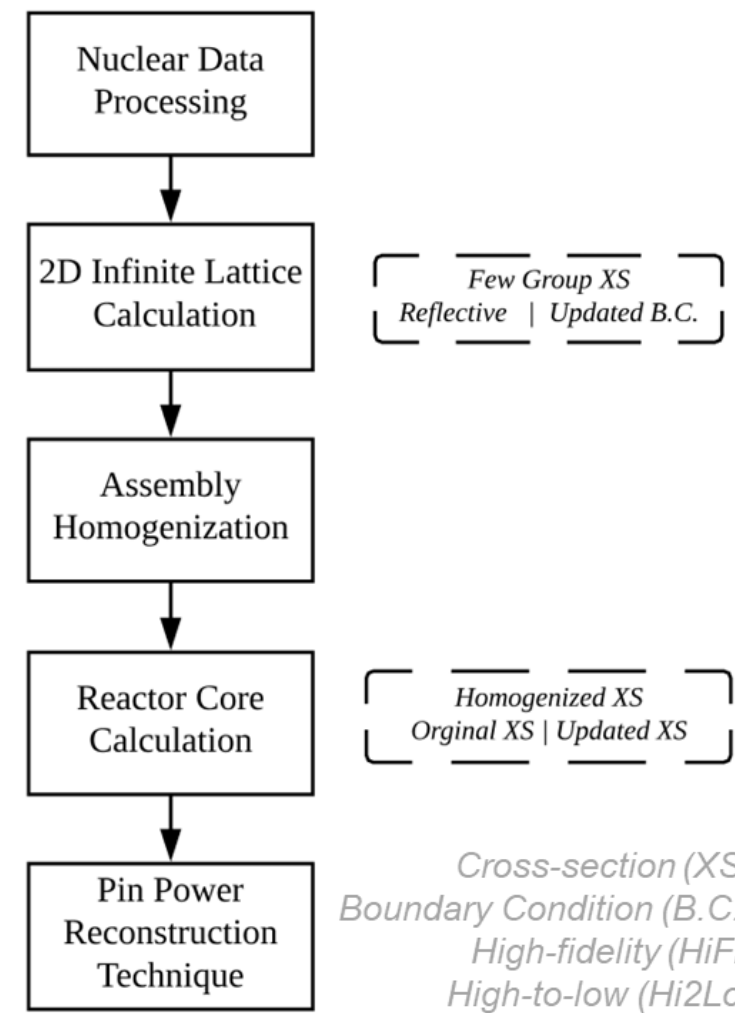
REACTOR PHYSICS Hi2Lo



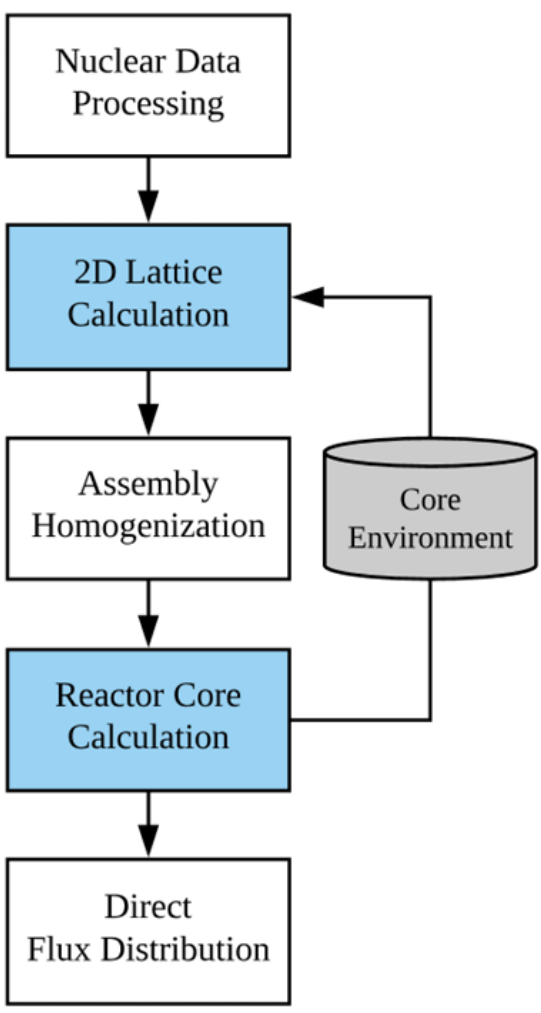
Full Core
Transport
Solution



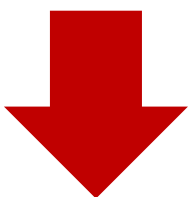
Two-step Method



Hi2Lo Method



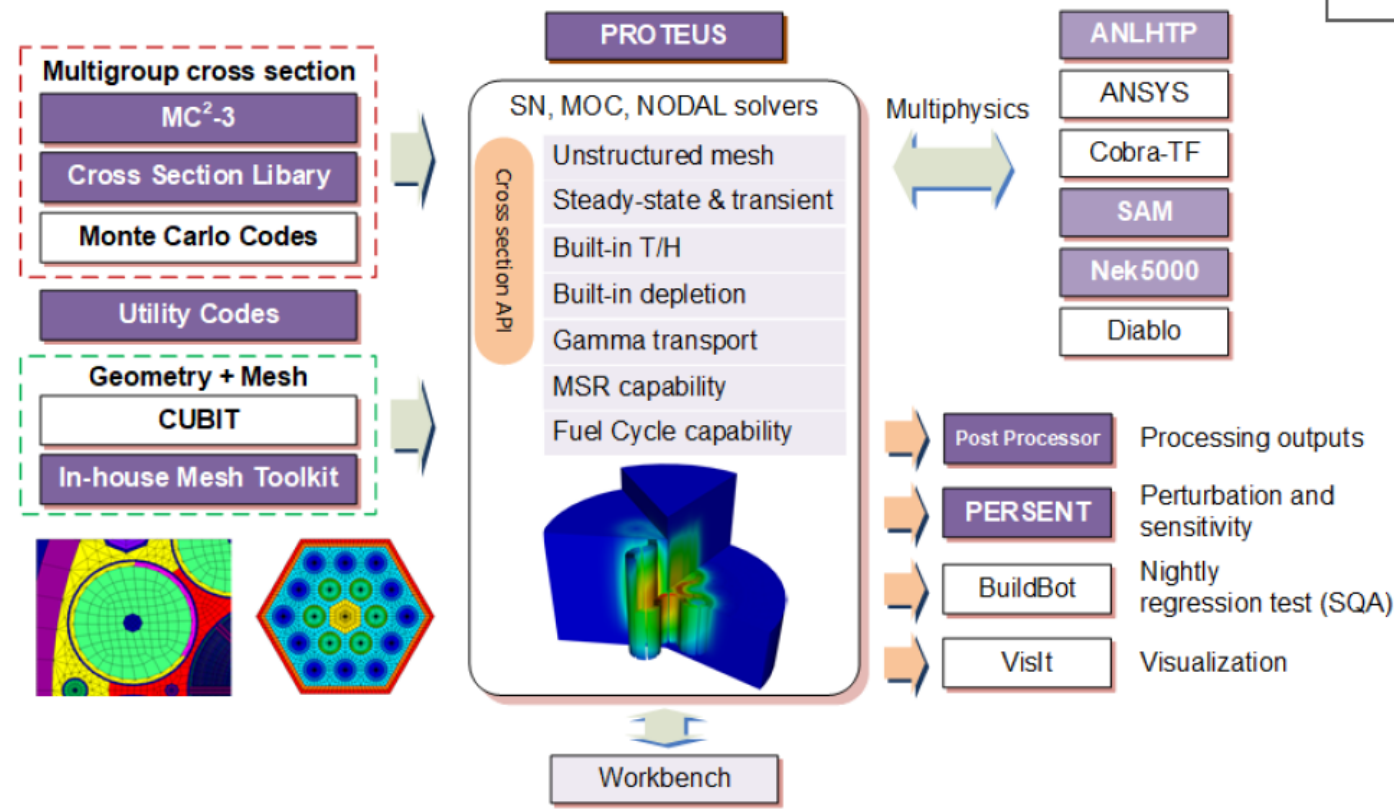
Two-step Method
Approximated core
environment
Approximated pin power



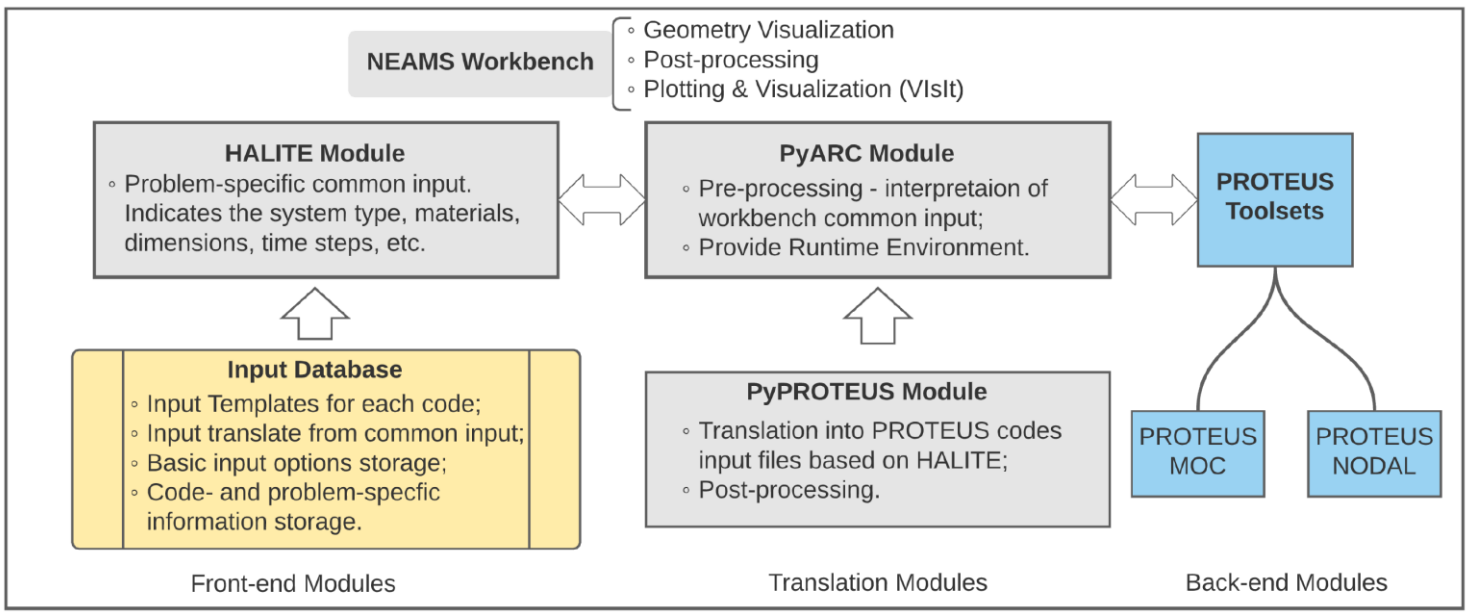
Hi2Lo Method
On-the-fly XS generation
Detailed flux distribution

REACTOR PHYSICS Hi2Lo

PROTEUS Code Package Integration into NEAMS

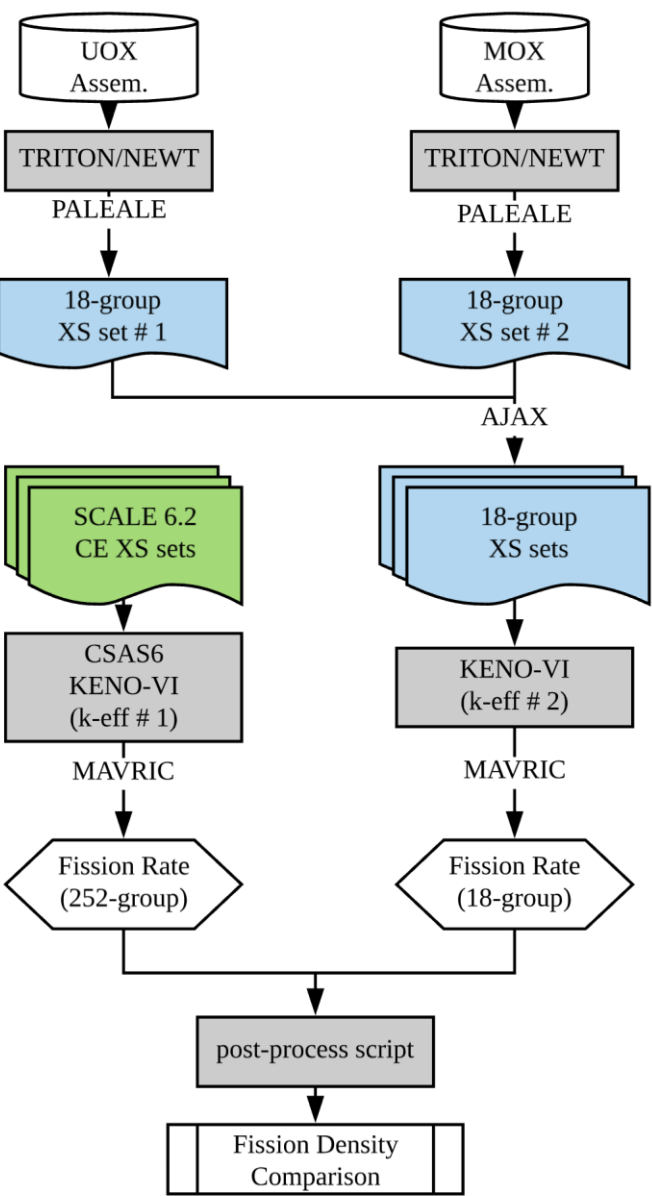


(The purple-colored boxes denote the codes developed by Argonne)



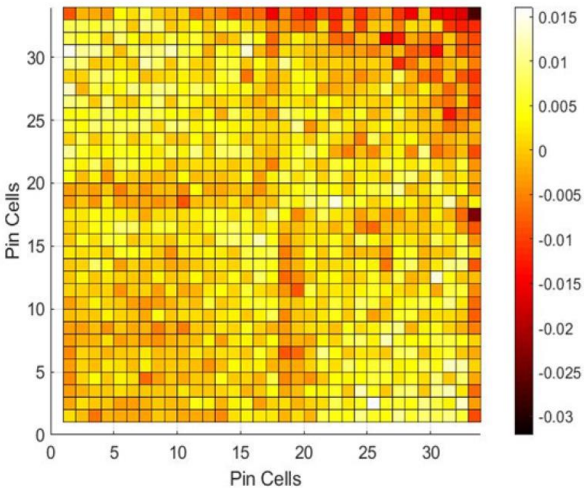
- A deterministic tool-sets that currently merged with Rattlesnake and become Griffin.
- 3 solvers: PROTEUS-SN, PROTEUS-MOC (MOCEX), PROTEUS-NODAL (NODAL)
- Lacks various capabilities for Hi2Lo and Multiphysics.

REACTOR PHYSICS Hi2Lo – XS LIBRARY GENERATION

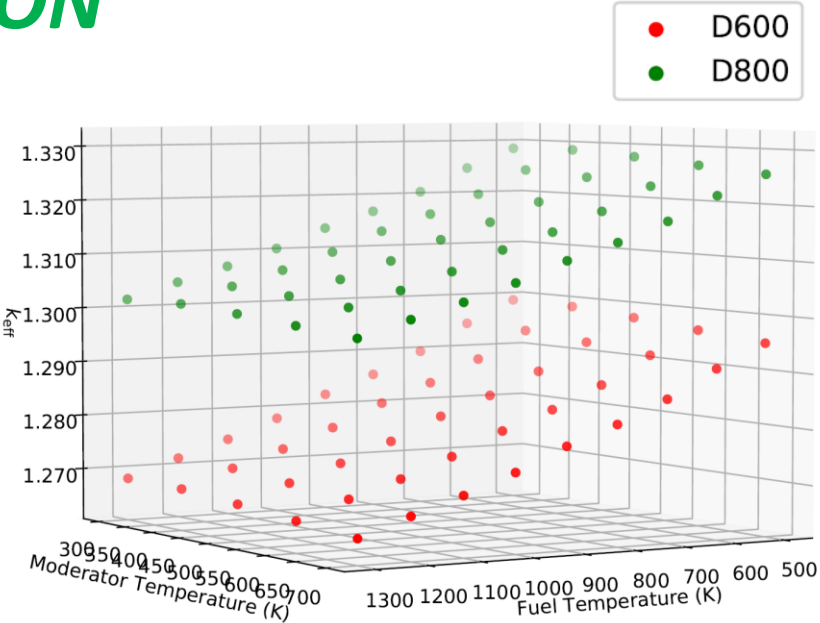


In Total: 225 sets of material-based XS

Nuclear Parameters		Analysis Value									
Fuel Temperature (K)		500	600	700	800	900	1000	1100	1200	1300	
Moderator Temperature (K)				300	400	500	600	700			
Moderator density (kg/m ³)				600	650	700	750	800			



FRD Error.



UOX k_{eff} of MD @ 600, 800 kg/m³

Cases/ Terms	k-eff	k-eff Difference (pcm)	FRD MIN/MAX Deviation (%)	FRD RMS (%)
CE (#1)	1.21438	97	-3.21/1.61	0.68%
18 Group (#2)	1.21341			

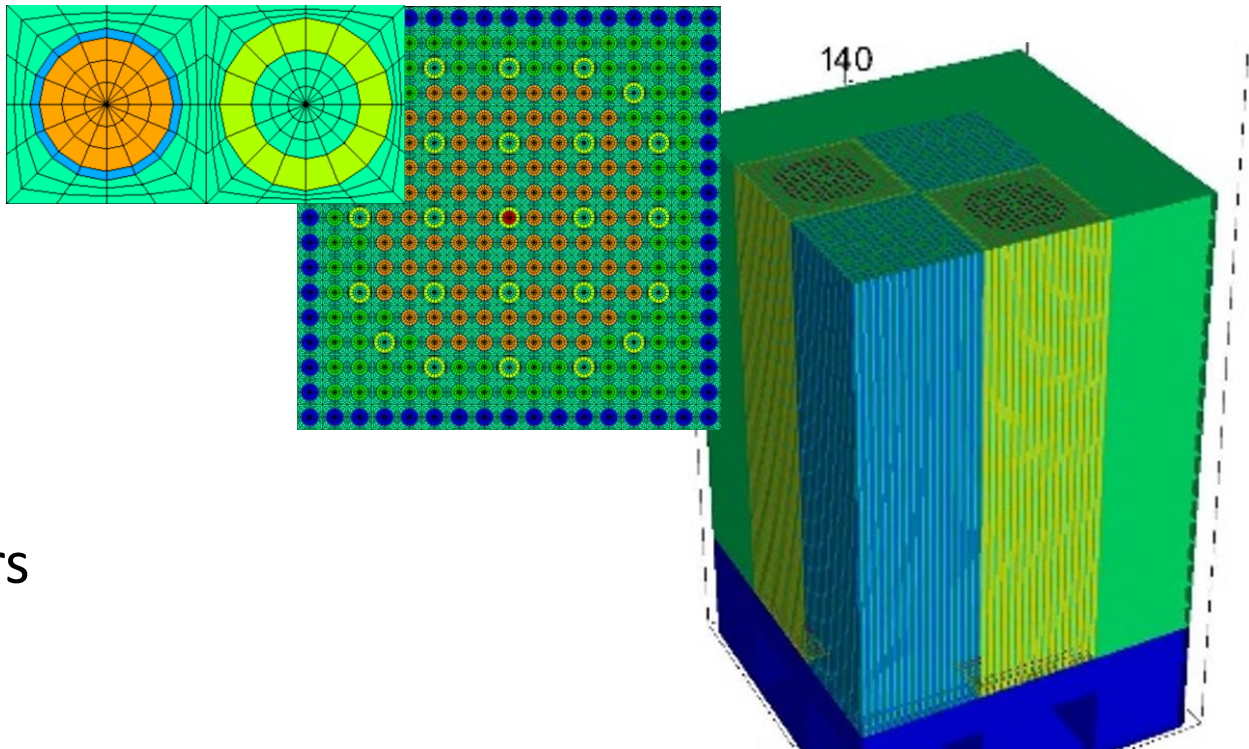
Observation:

- Good agreement between CE and 18 groups results
- Larger difference near reflector - normal effect in group condensation
- Reflect a correct physics behavior

REACTOR PHYSICS Hi2Lo – REFERENCE SOLUTION

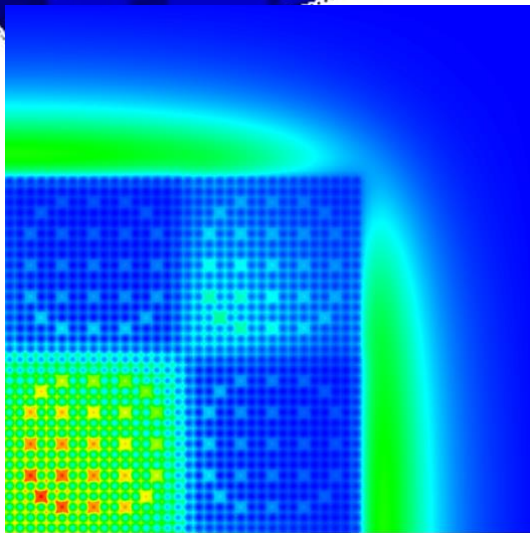
Full core transport calculation with MOC model:

- 96 angles
- Convergnce criteria 10^{-6} for k_{eff} , source, flux
- 30,193 spatial meshes & 296,736 trajectories per assembly
- 16 azimuthal angles and 16 surface mesh sectors per pin



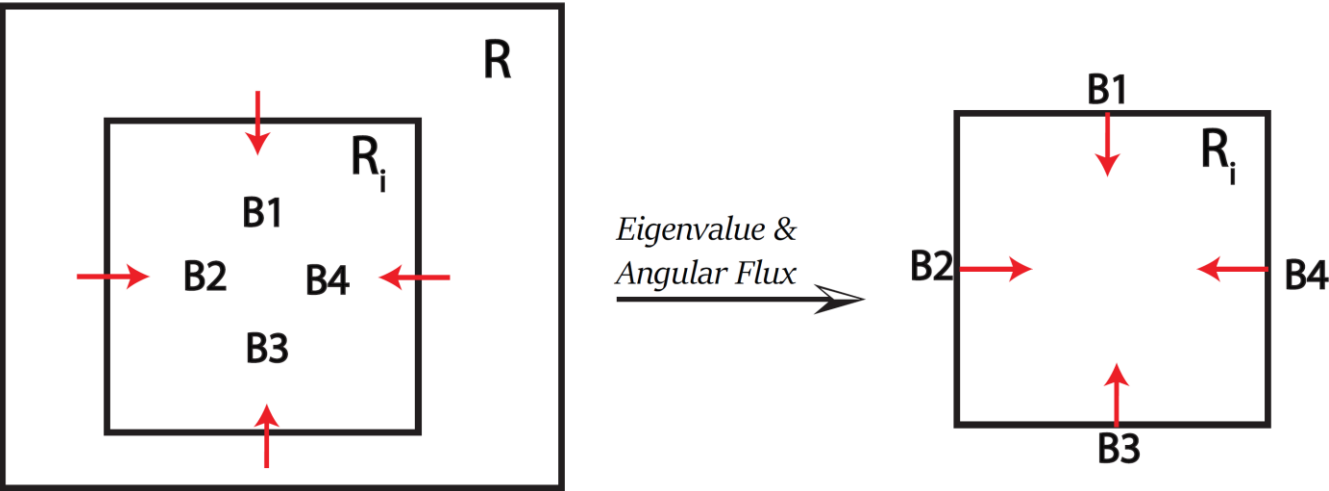
2D MOCEX result is verified against KENO-VI, order to order

Scattering order	KENO-VI[std.] (Ref.)	Difference (pcm)
P5	1.21316[10]	-38.82
P3	1.21346[10]	48.92
P1	1.21294[10]	42.53
P0	1.24741[10]	2.25



MOCEX P0 is used as reference in the following calc

REACTOR PHYSICS Hi2Lo – FIXED K_{EFF} ALGORITHM



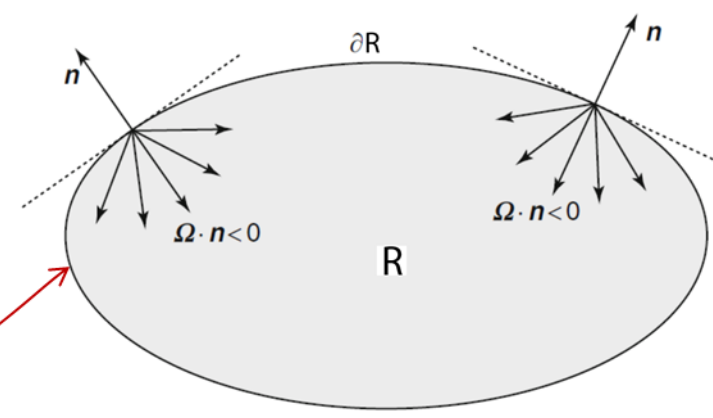
- Decouple the full domain calculation into individual sub-domain calculations.
- Solved in parallel and retains the local physical response: flux distribution, leakage, ...
- This can only be done iteratively.

$$\Omega \cdot \nabla \psi_i(\mathbf{r}, E, \Omega) + \Sigma_{t,i}(\mathbf{r}, E) \psi_i(\mathbf{r}, E, \Omega) =$$

$$\int_0^\infty \int_{4\pi} \Sigma_{s,i}(\mathbf{r}, E' \rightarrow E, \Omega' \cdot \Omega) \psi_i(\mathbf{r}, E', \Omega') dE' d\Omega'$$

$$+ \frac{\chi_p(\mathbf{r}, E)}{4\pi k_m} \int_0^\infty \int_{4\pi} \nu \Sigma_f(\mathbf{r}, E') \psi_i(\mathbf{r}, E', \Omega') dE' d\Omega', \mathbf{r} \in R_i, \Omega \in 4\pi$$

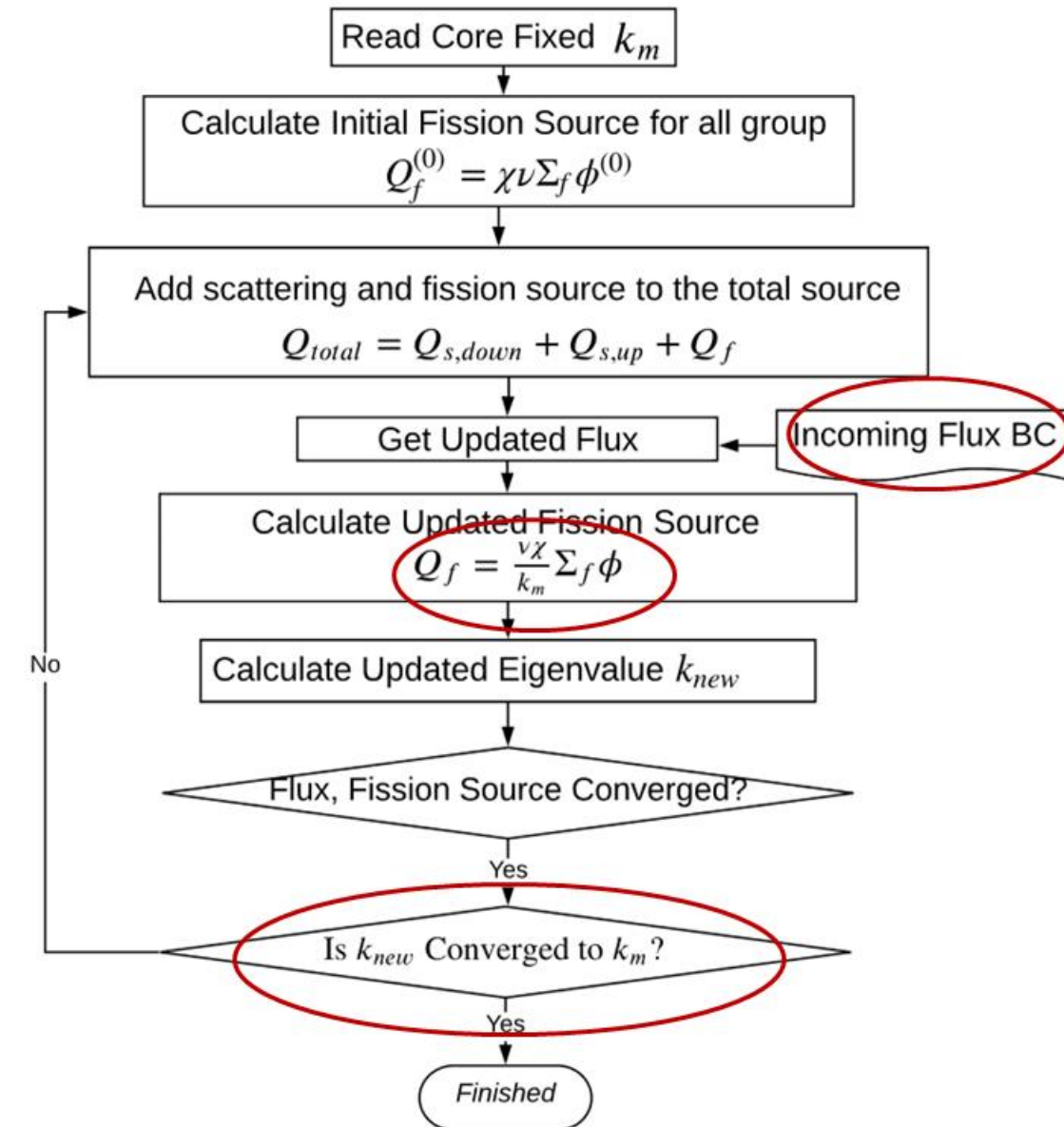
$$\psi_i^b(\mathbf{r}, E, \Omega) = \psi(\mathbf{r}, E, \Omega), \mathbf{r} \in \partial R_i, \Omega \cdot \vec{n} < 0.$$



System with non-trivial solution exists

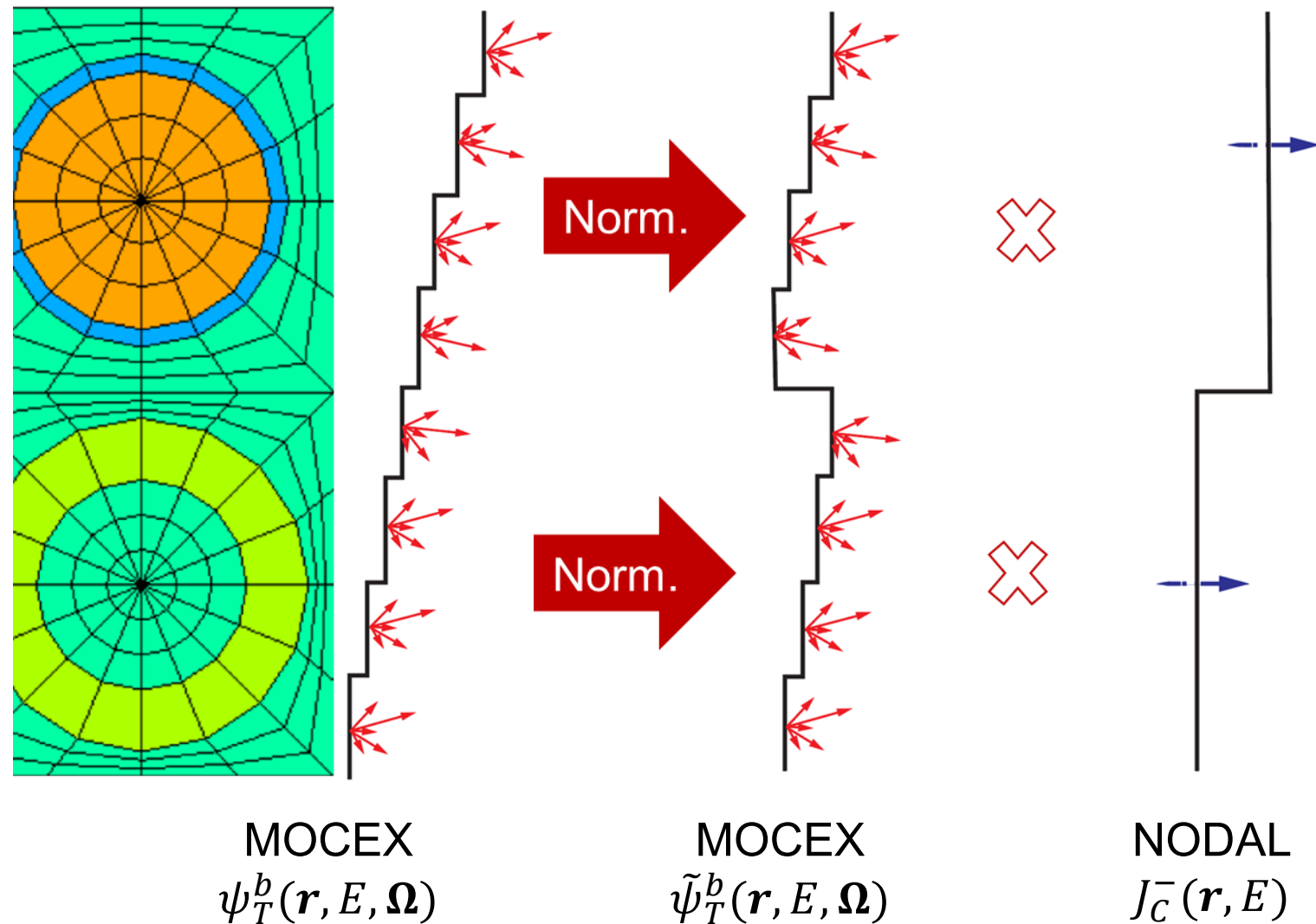
REACTOR PHYSICS HI2LO – FIXED k_{EFF} ALGORITHM

- The two main neutronic types of calculations are: (1) the fixed source or (2) the critical eigenvalue.
- Here a different type of calculation is needed where the eigenvalue is fixed at the value of the core calculation, and the goal is to find the corresponding neutron flux within an assembly calculation where the boundary conditions are known.
- The resulting flux is used to condense the multi-group cross-sections.
- Three major code modifications in MOC calculations for generating the cross-sections:
 1. Inner iteration BC operation
 2. Fission source update module
 3. Convergence checking module



REACTOR PHYSICS Hi2Lo – ANGULAR BOUNDARY CONDITIONS

- The available information from the core calculation regarding boundary conditions is the partial current on a coarse surface.
- However, the MOC calculation requires the inter-assembly angular flux (IAAF) on the fine surface boundaries. A specific reconstruction approach is used.



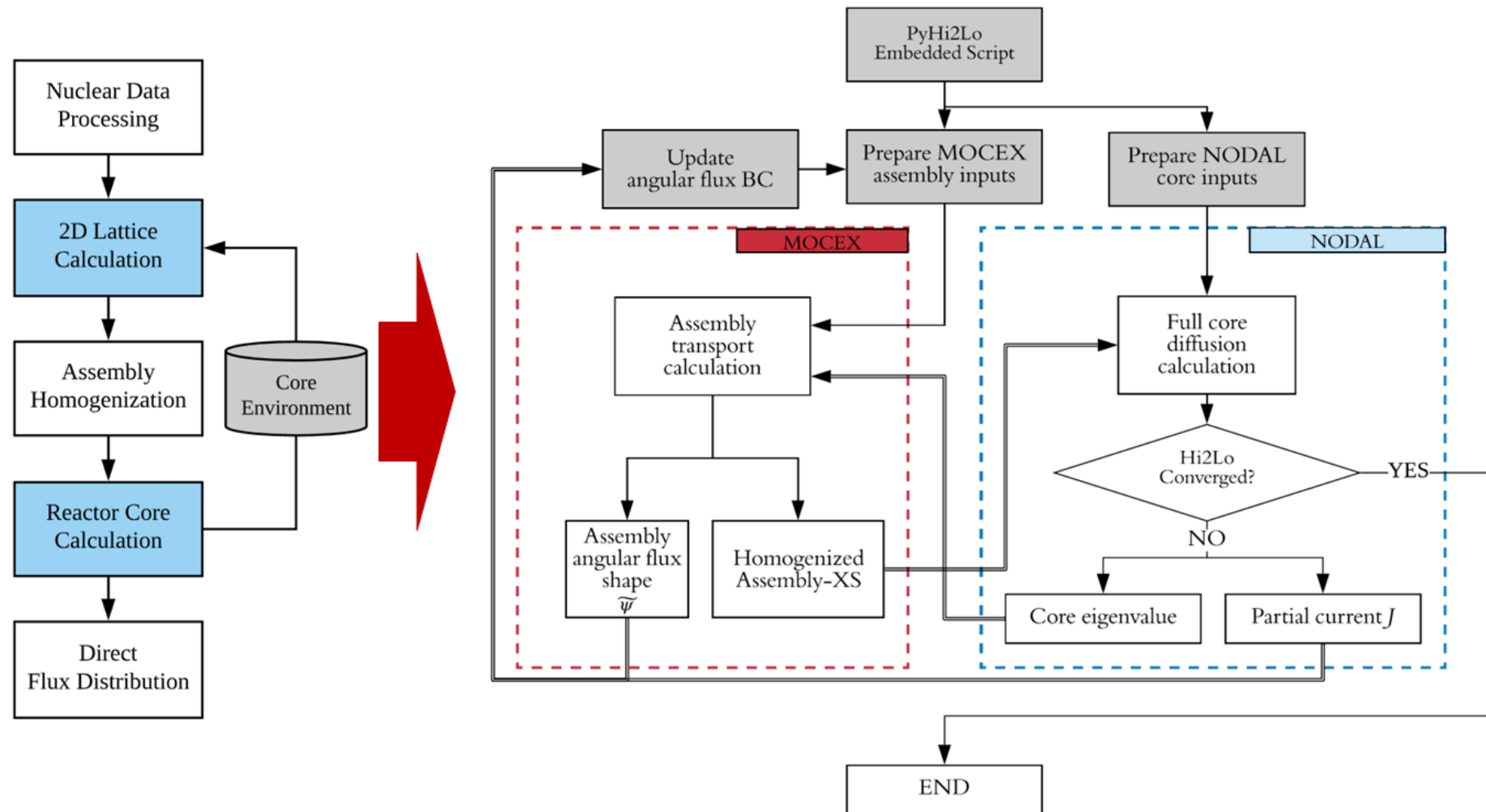
- **Norm.:** Angular flux Normalization

$$\tilde{\psi}_T^b(\mathbf{r}, E, \mathbf{\Omega}) = \frac{\psi_T^b(\mathbf{r}, E, \mathbf{\Omega})}{\int_l dr \int_0^{2\pi} d\Omega \psi_T^b(\mathbf{r}, E, \mathbf{\Omega})}$$

- **X:** Partial current multiplication

$$\psi_T^b(\mathbf{r}, E, \mathbf{\Omega}^-) = \tilde{\psi}_T^b(\mathbf{r}, E, \mathbf{\Omega}) \times J_C^-(\mathbf{r}, E)$$

REACTOR PHYSICS Hi2Lo – ITERATIVE SCHEME



Global Convergence criteria

1. Core k_{eff} change:

$$\delta k_C^n = \left| \frac{k_C^n - k_C^{n-1}}{k_C^{n-1}} \right| < \epsilon_k$$

2. Pin power change:

$$\delta F_T^n = \left| 1 - \frac{F_T^n}{F_T^{n-1}} \right| < \epsilon_F$$

- ϵ_k selected as 1.0×10^{-5}
- ϵ_F selected as 1.0×10^{-3}

REACTOR PHYSICS Hi2Lo – RESULTS

MOCEX settings:

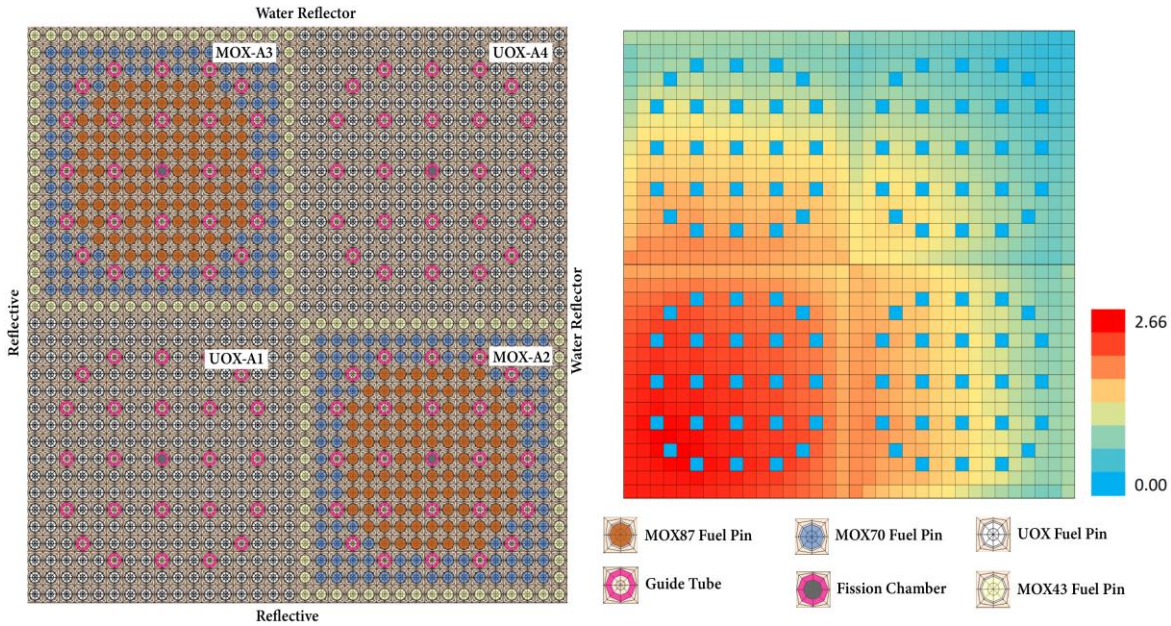
- 96 angles
- Convergence criteria $10^{-6}/10^{-6}/10^{-6}$ for k_{eff} , source, flux
- 30,193 spatial meshes & 296,736 trajectories per assembly

NODAL settings:

- Convergence criteria $10^{-6}/10^{-6}/10^{-6}$ for k_{eff} , source, flux
- Receive assembly homogenized 18Group XS
- One-node-per-pin calculation
- No DF used

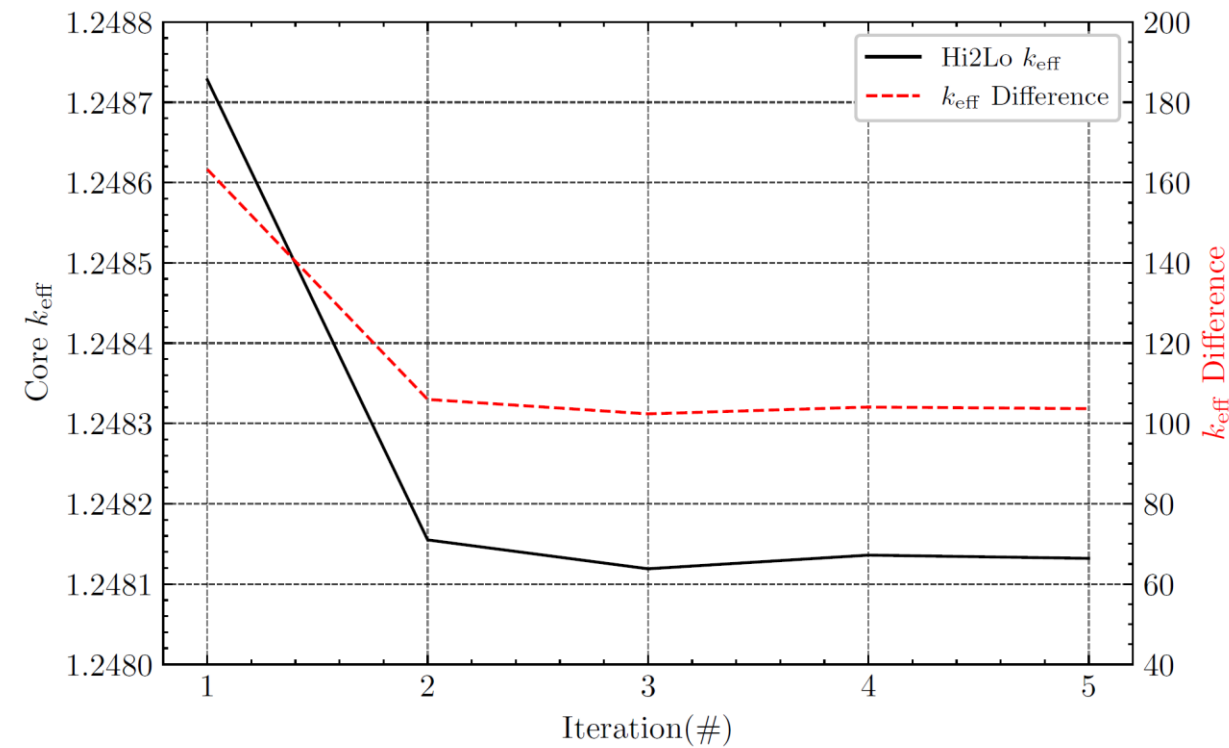
Case	Core k_{eff}	diff. (pcm)	Wall-time (min)
Reference MOCEX	1.24710	–	2076.4
Reference NODAL	1.24797	87.4	2078.4
Off-line	1.24873	163.3	19.6
Hi2Lo	1.24813	103.7	1473.9

*Reference NODAL: core calculation uses location-dependent assembly XS from reference calculation.

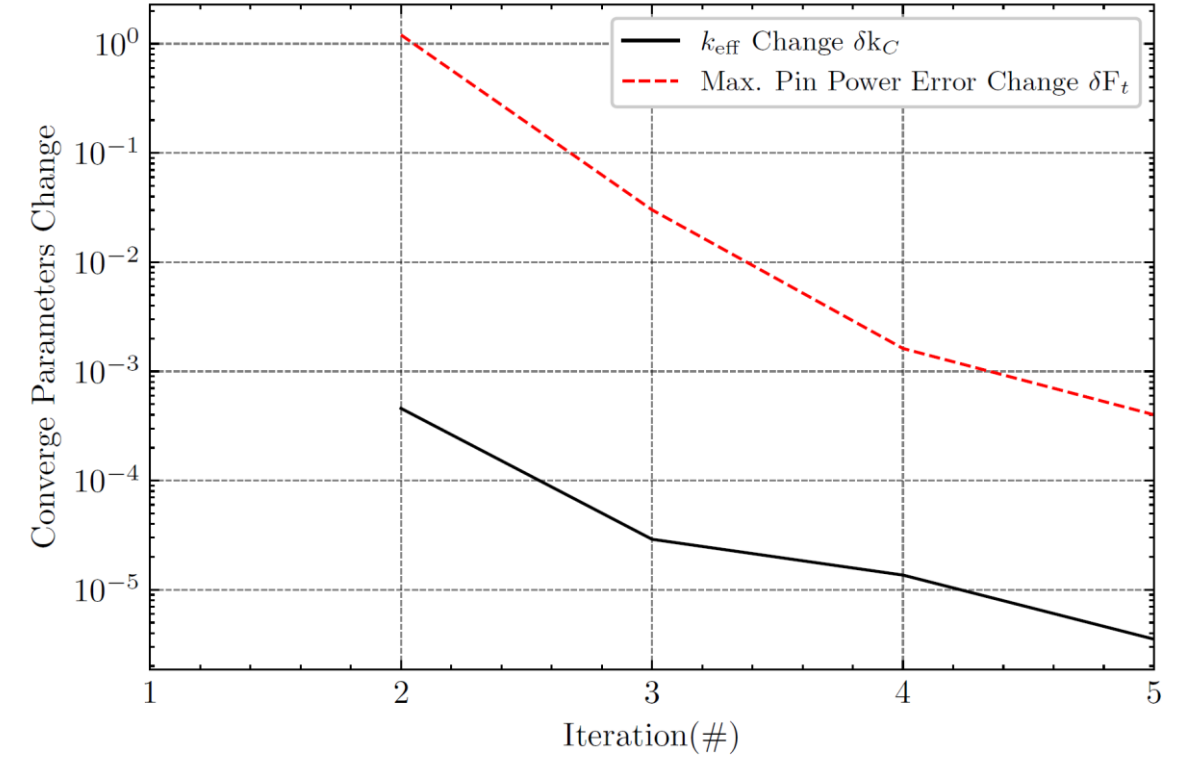


- k_{eff} result improved by ~60 pcm
- k_{eff} result improved towards the reference NODAL
- Calculation time shorten by ~40%

REACTOR PHYSICS Hi2Lo – CONVERGENCE



Eigenvalue

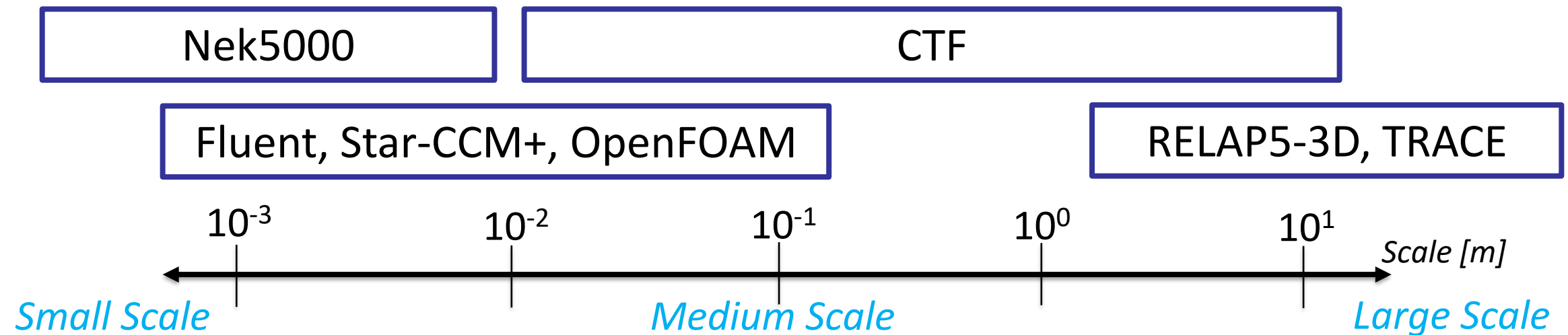


Convergence parameters

- Major improvement of k_{eff} happens at the first several iterations
- The k_{eff} error is self-corrected through global iterations
- The k_{eff} and pin power change decrease with 1 order of magnitude per global iteration

THERMAL-HYDRAULICS Hi2Lo

- The main modeling aspects that differ from high to low fidelity thermal-hydraulics models are:
 - Small (near surfaces) vs medium (subchannel) vs large (assembly) scales.
 - Cross-flows
 - Turbulent phenomena
- Examples of codes fidelity:



+ ← Resolution → -

- ← Scale → +

- ← Assumptions → +

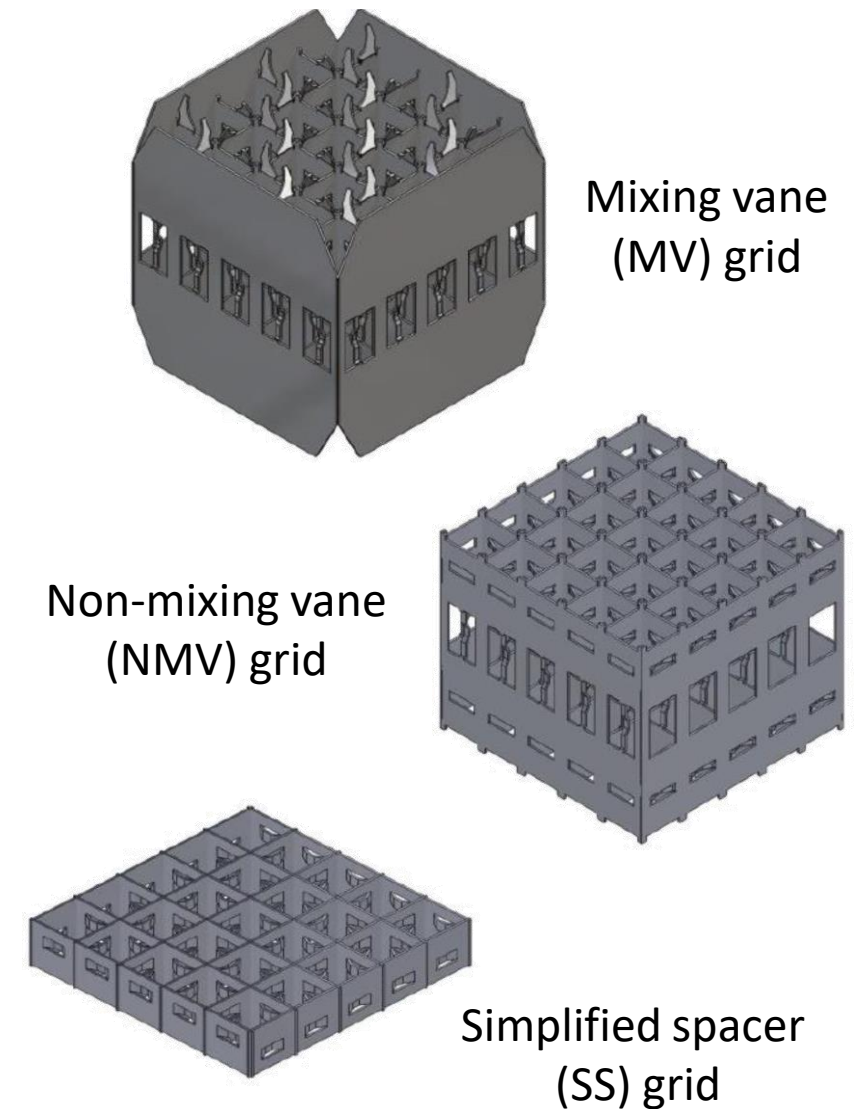
THERMAL-HYDRAULICS Hi2Lo

- The Hi2Lo approach aims at informing a low fidelity code (CTF) based on high fidelity simulations (NEK5000).
- An offline approach is used to improve:
 - Pressure losses around spacer grids
 - Subchannel friction factor
- C5G7 core is used as an example.

Thermal-Hydraulics Hi2Lo – Spacer Grid Pressure Loss

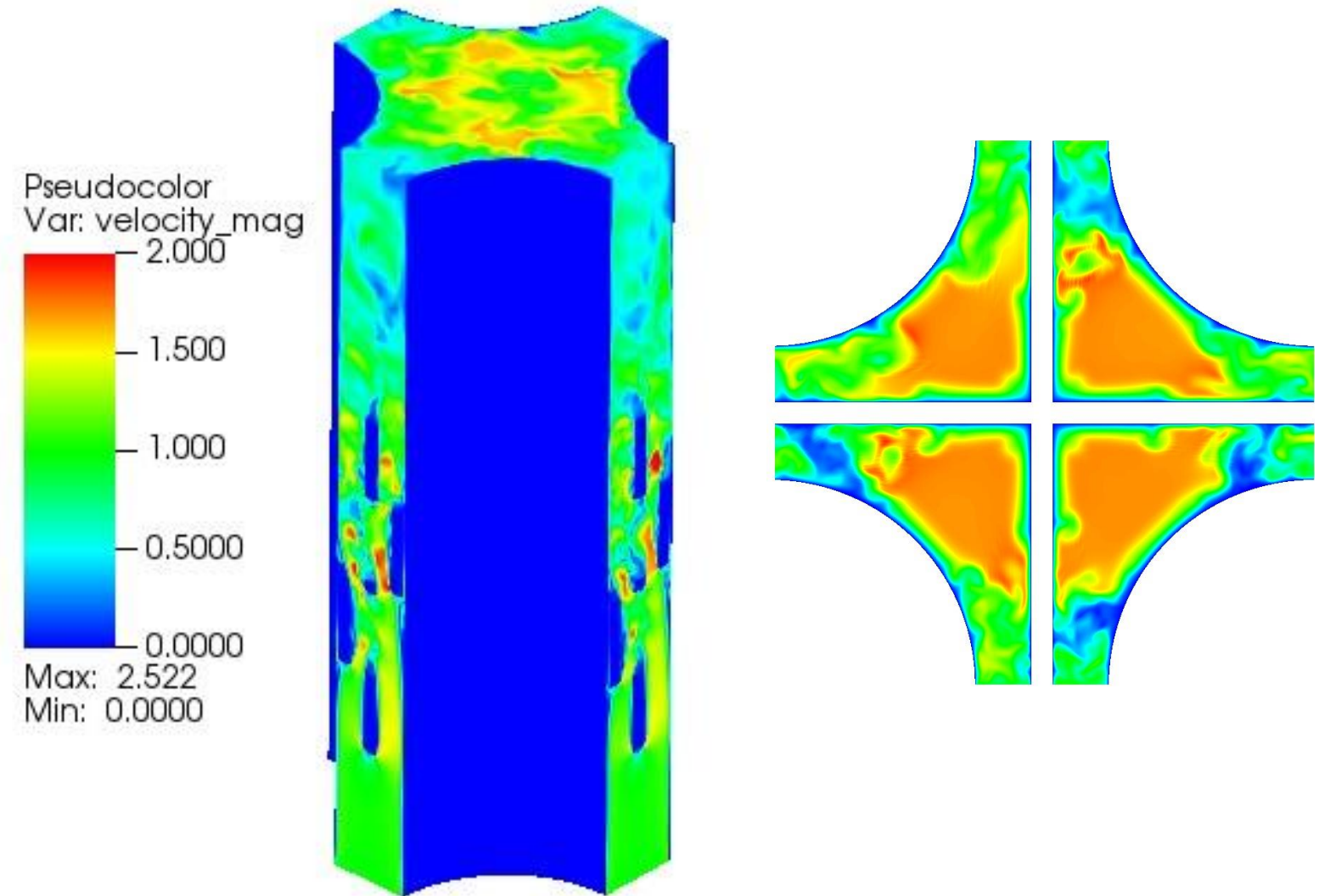
- Inform CTF with total grid spacer loss coefficient from Nek5000 solution
 - Nek5000 yields the non-dimensional pressure drop (P^*) and velocity (u^*) by solving the non-dimensional Navier Stokes equations
 - Total grid loss coefficient (C_{DL}) is defined as:

$$C_{DL} = 2\Delta P^*$$



THERMAL-HYDRAULICS Hi2Lo – SPACER GRID PRESSURE LOSS

- Pressure loss is computed as the difference between two planes before and after the spacer grids.



THERMAL-HYDRAULICS Hi2Lo – SPACER GRID PRESSURE LOSS

- Two methods of averaging were investigated
 - Thin-volume (points within a thin volume)
 - History points (interpolated probe points)
- Both methods agree well for most geometries

Geometry	Reynolds Number	T_{FINAL}	History Points C_{DL}	Thin Volume C_{DL}	Error Percentage (%)
MV corner	1,000	10.0	5.648	5.658	0.176
MV corner	10,000	4.8	2.305	2.348	1.405
NMV corner	1,000	2.9	3.308	3.773	12.326
NMV center	1,000	1.6	4.757	4.805	0.998
MV center	1,000	1.0	5.074	5.117	0.841

THERMAL-HYDRAULICS Hi2Lo – SUBCHANNEL FRICTION FACTOR

- Inform CTF with bare subchannel friction factor from Nek5000 solution

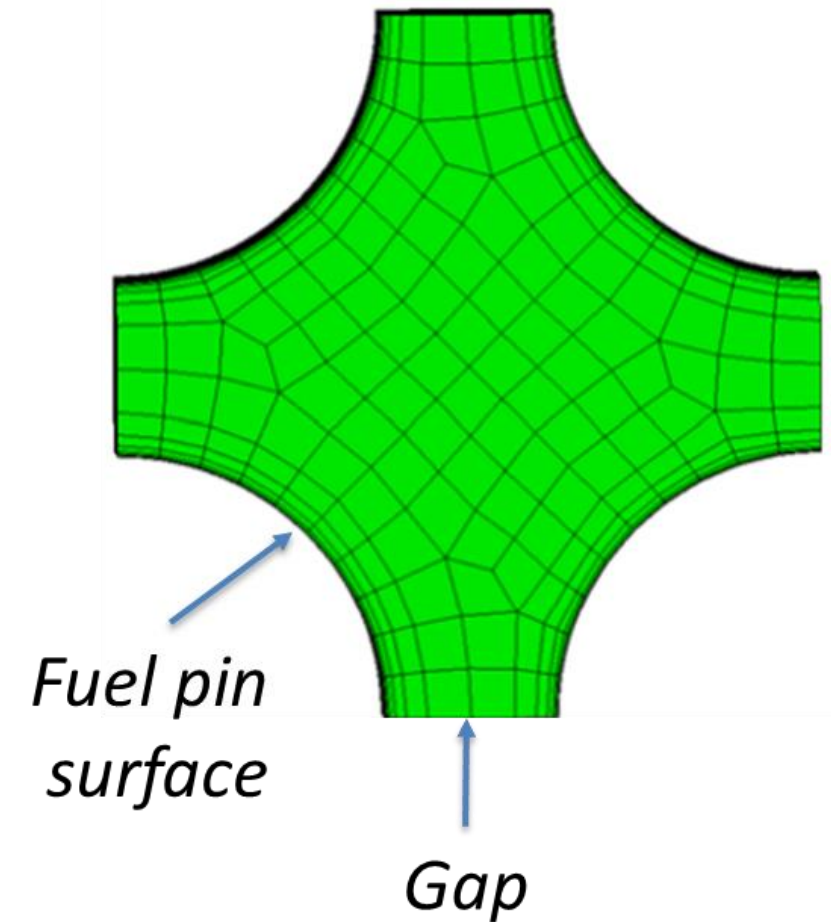
- Friction factor in CTF (f_{CTF}) is defined as:

$$f_{CTF} = A + B * Re^C$$

- Friction factor from Nek5000 is calculated using the Colebrook equation:

$$f^{-0.5} = -2 \log \left(\frac{\epsilon}{3.7D} + \frac{2.51}{Re f^{0.5}} \right)$$

- Three polynomial orders (P = 5,7,9) were investigated. Polynomial order is required for spectral element solver in Nek5000



THERMAL-HYDRAULICS Hi2Lo – SUBCHANNEL FRICTION FACTOR

- Friction velocity (left table) was obtained from Nek5000 for three polynomial orders.
- CTF friction factor coefficients (right table) were calibrated for three Nek5000 polynomial orders.
- Uncertainty due to the grid size can be propagated to CTF through the coefficients.

	Polynomial Order		
Reynolds Number	P = 5	P = 7	P = 9
8,000	0.035382	0.035383	0.035440
9,000	0.033351	0.033486	0.033426
10,000	0.031684	0.031660	0.031690
11,000	0.030212	0.030161	0.030101
12,000	0.028956	0.028987	0.028972

Polynomial Order	A	B	C
P = 5	0.0	18.797	-0.993
P = 7	0.0	17.327	-0.984
P = 9	0.0	19.079	-0.994

FUEL PERFORMANCE Hi2Lo

High Fidelity:

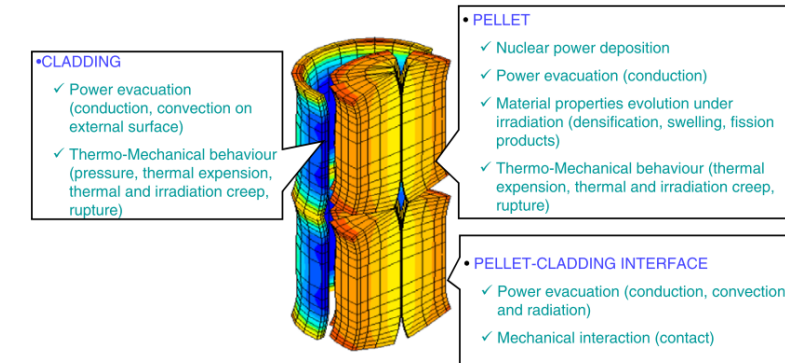
- Finite element or finite volume method 3D/2D or 1.5D.
- Full coupling of thermal, mechanical and physico-chemical aspects.
- Models for most important phenomena but computationally expensive
- Examples: BISON, OFFBEAT, ALCYONE.

Standard Fidelity:

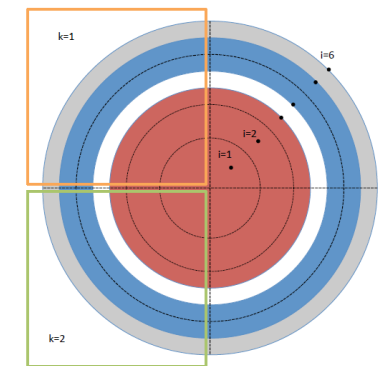
- Finite difference 1.5D.
- Full coupling of thermal, mechanical and physico-chemical aspects.
- Models for most important phenomena and computationally efficient.
- Examples: FAST/FRAPCON/FRAPTRAN, TRANSURANUS

Low Fidelity:

- Finite difference 1D.
- Solve only heat conduction in fuel and cladding by imposing the gap heat transfer. Few thermal phenomena are considered.
- Computationally very cheap and usually consist thermal solvers of thermal-hydraulics codes.
- Examples: CTFFuel, FINIX



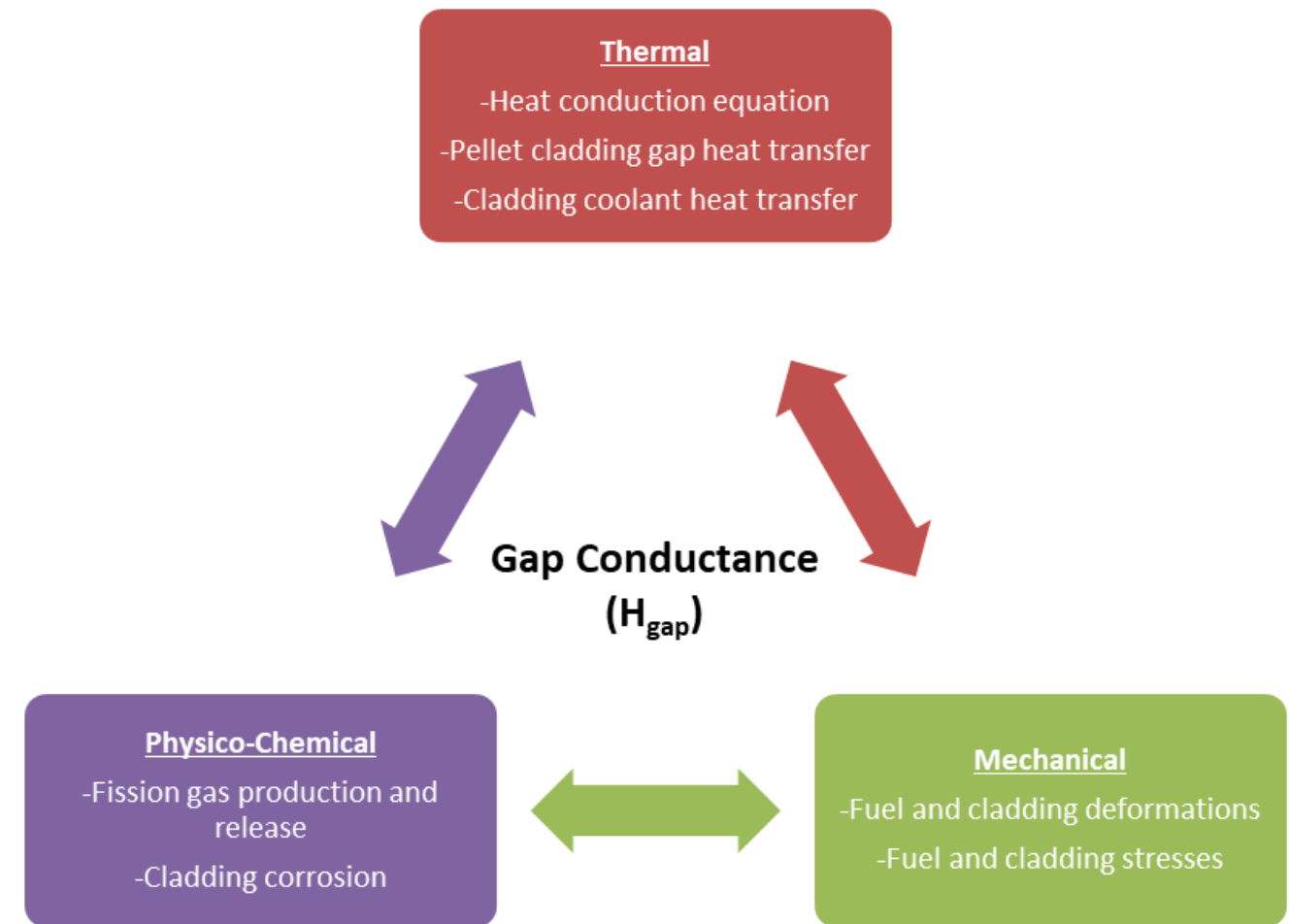
*<http://www.materials.cea.fr/en/PDF/MonographiesDEN/Nuclear-fuels-CEA-en.pdf>



*R. Salko et al., CTF Theory Manual, 2019

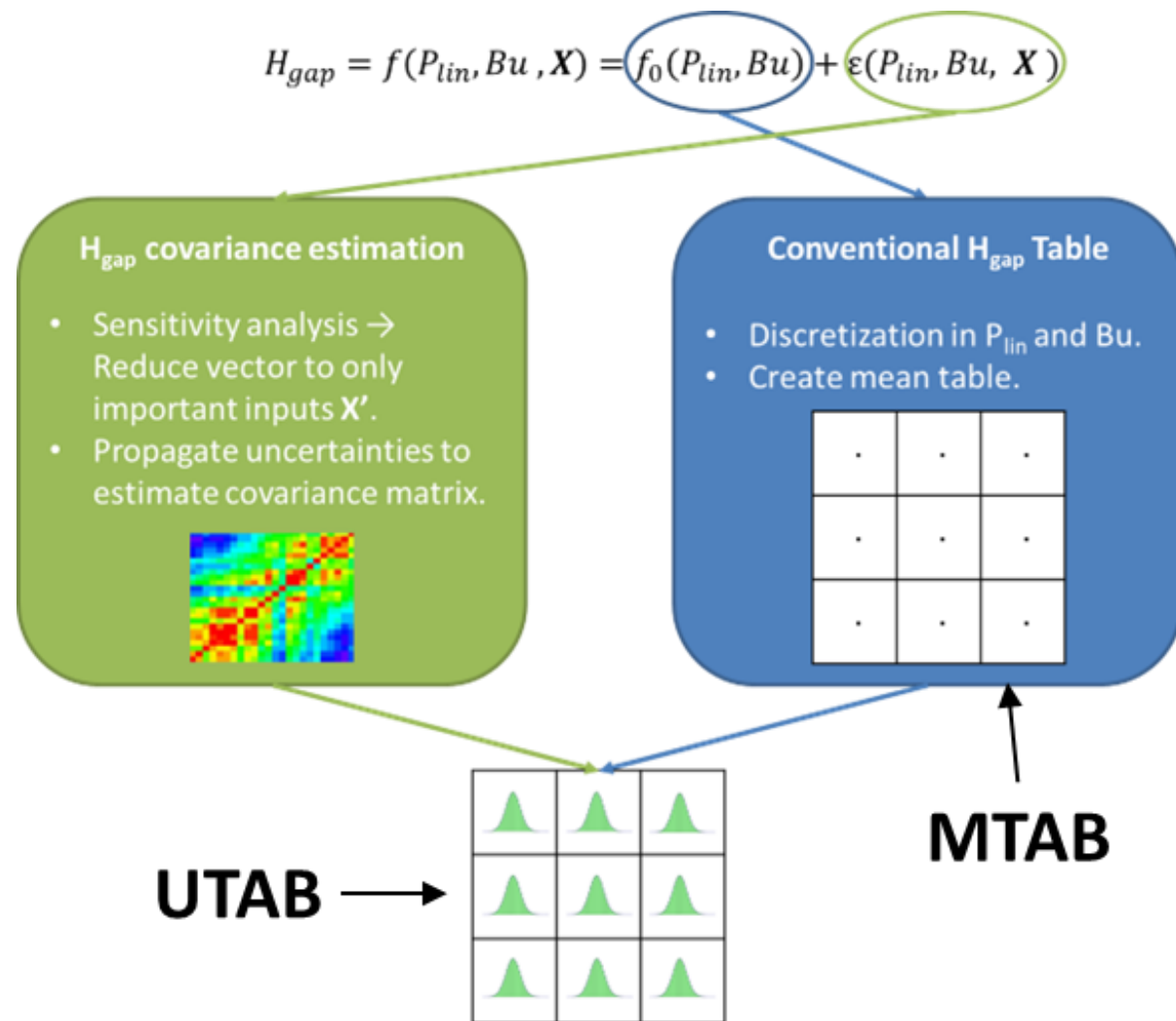
FUEL PERFORMANCE Hi2Lo – GAP CONDUCTANCE

- Pellet cladding gap conductance (H_{gap}) is selected as the fuel performance Hi2Lo quantity.
- H_{gap} is a good candidate because it is a result of the high-fidelity fuel performance codes and an input to the low fidelity thermal solvers.
- Within H_{gap} many high fidelity fuel performance models are contributing that are not modelled in low fidelity thermal solvers. Examples are fission gas release and creep.
- This allows for an easier uncertainty propagation of high-fidelity fuel performance models.



FUEL PERFORMANCE Hi2Lo – UTAB

UTAB – Uncertain Tabular approach



- Approach similar to multigroup multiparametric cross-sections in neutronics.
- Extension of conventional tabulated approach (**MTAB**) constructed based on a high-fidelity code for different condition parameters (e.g. linear power and burnup).
- Screening of inputs to identify the important ones for H_{gap} during depletion.
- **MTAB** is used as the mean table and for each grid point the associated covariances due to remaining influential uncertain inputs are estimated using the high-fidelity code.
- **UTAB** samples of the inputs and the consistent H_{gap} tables are provided to the low fidelity code.

FUEL PERFORMANCE Hi2Lo – UTAB

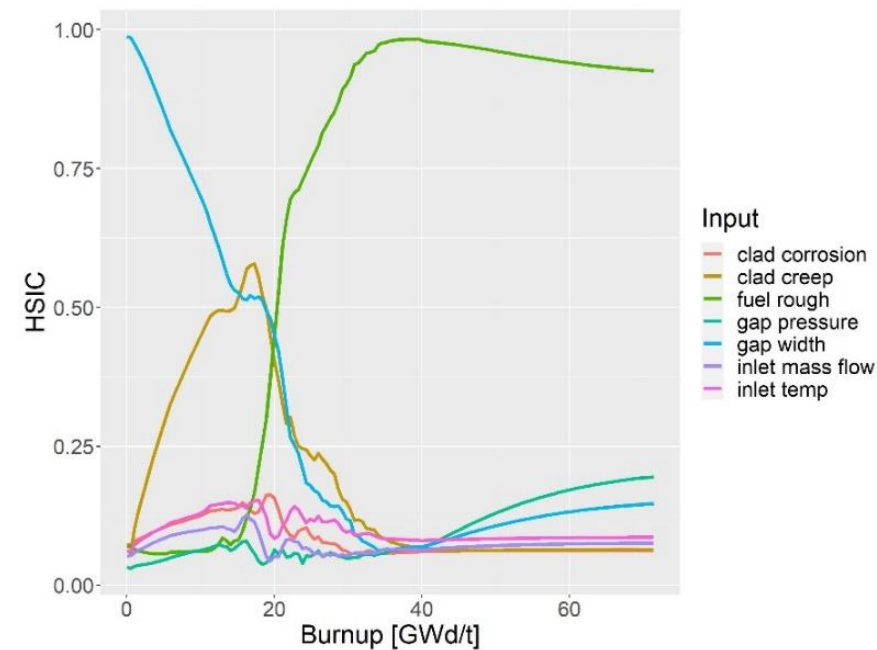
- Two different approaches were identified for the depletion calculations:
 1. VARDEP (varying depletion) : For each grid point the linear power is used as constant for the whole depletion.
 2. REFDEP (reference depletion): For each grid point the depletion is always performed with a reference linear power ($20 \frac{kW}{m}$) and a step jump is performed at the linear power of each grid point.
- The H_{gap} value at the mid-height location is used for UTAB if more than one axial slice is modelled.
- FRAPCON and BISON were used to inform CTFFuel
- Example developed for C5G7 fuel rods (UOX and MOX).

FUEL PERFORMANCE Hi2Lo – UTAB

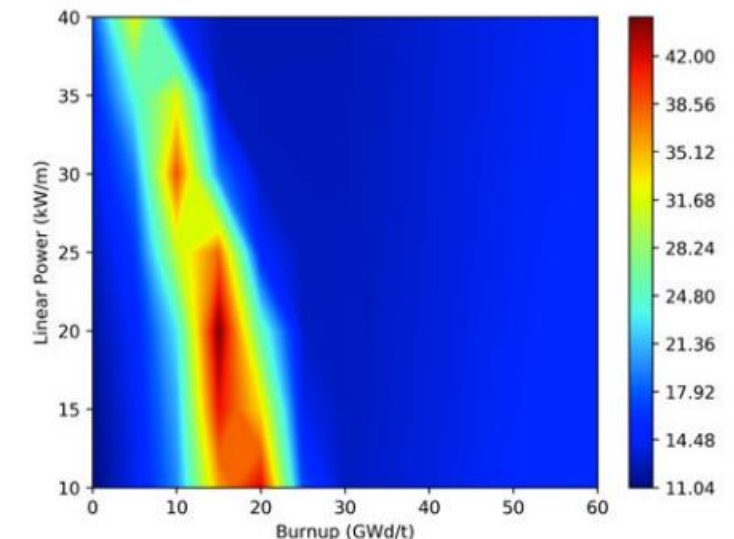
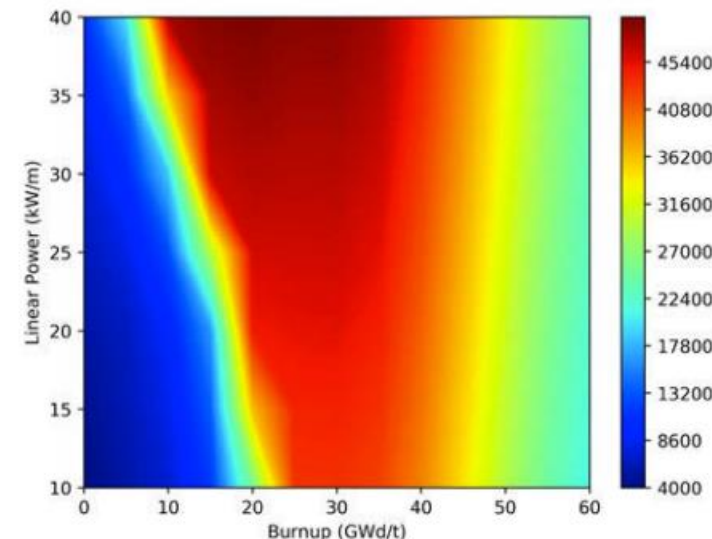
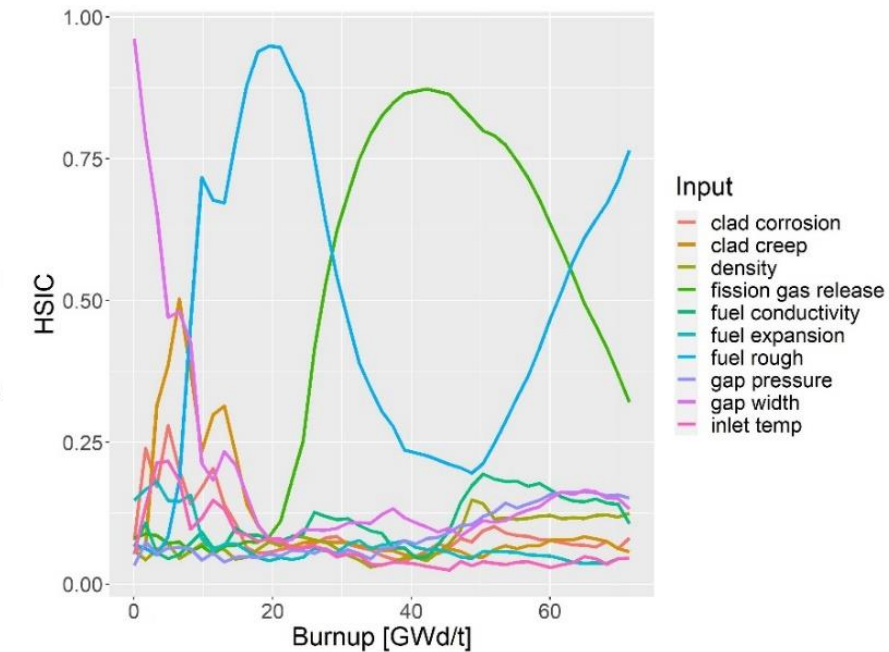
- Linear power and burnup are used as condition parameters.
- Screening using HSIC indices* calculated with FRAPCON:
 - Gap width
 - Fuel roughness
 - Cladding creep
 - Fission gas release
 - Grain growth
- FRAPCON calculated uncertainty increases in the region of linear power and burnup due to the sharp discontinuity of the gap closure.

*B. looss and A. Marrel, "Advanced methodology for uncertainty propagation in computer experiments with large number of inputs," Nuclear Technology, pages 1-19, 2019.

Depletion at 10kW/m



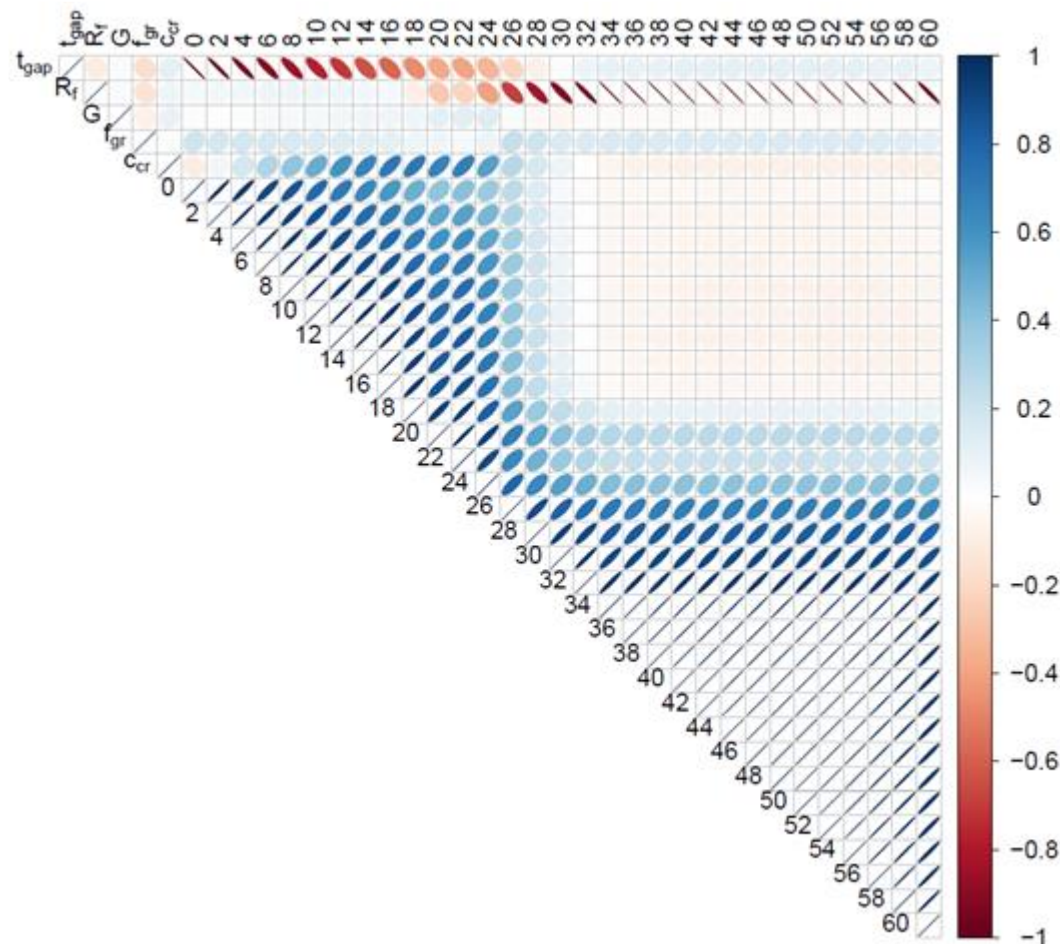
Depletion at 30kW/m



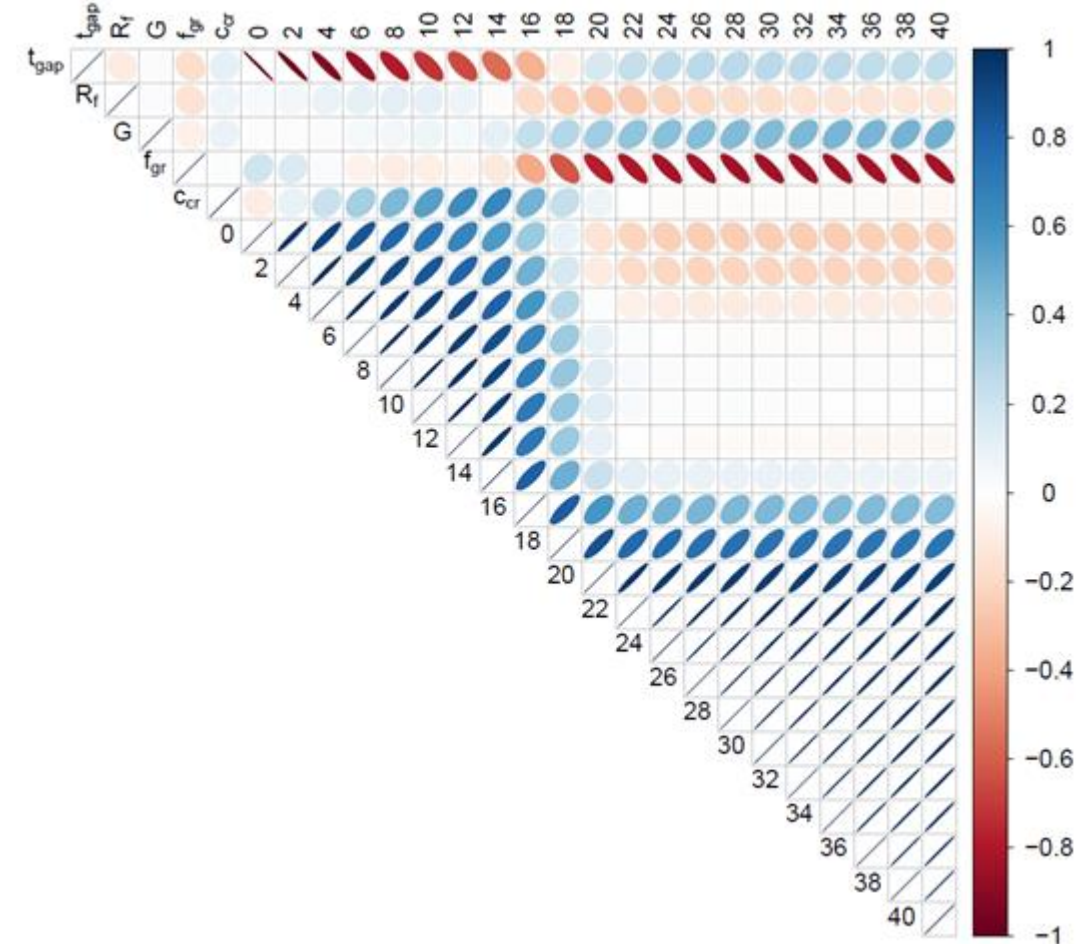
FUEL PERFORMANCE Hi2Lo – UTAB

- The correlation matrices indicate three different clusters of H_{gap} grid points: one before closure due to the gap width and creep, one after closure due to the fuel roughness and one at high burnup due to fission gas release.

Depletion at 10kW/m

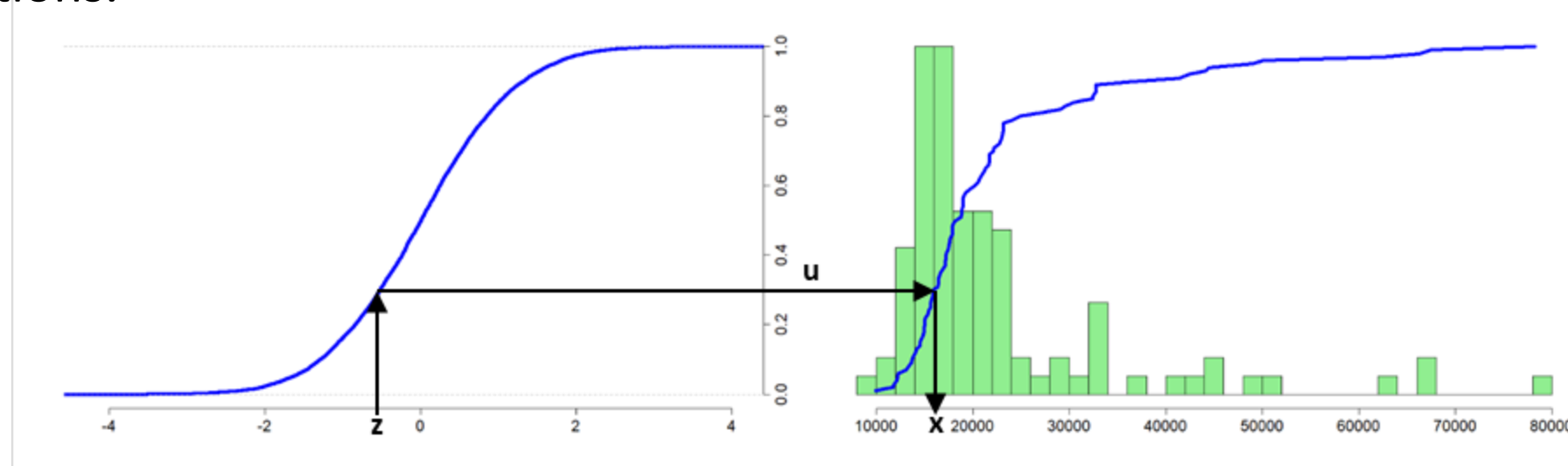


Depletion at 30kW/m



FUEL PERFORMANCE Hi2Lo – UTAB

- Cross-sections are easy to sample because they are assumed to be multivariate normal. However, this is not the case for gap conductance parametric values.
- A copula based approach is used to be able to sample from non multivariate normal distributions.

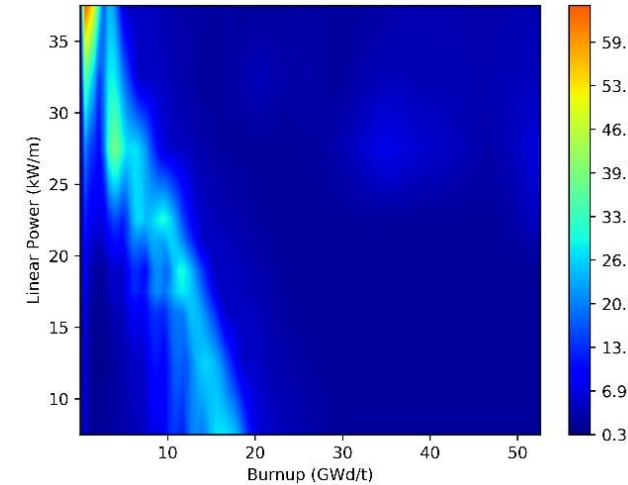


1. From UTAB we estimated the correlation matrix R (better use Spearman coefficients to account for H_{gap} non-linearity).
2. Generate M samples of $z \sim N(0, R)$, $z \in R^d$. Where d is the number of screened inputs + number of H_{gap} grid points.
3. Compute marginal quantiles $u_1 \dots u_d = F(z_1) \dots F(z_d)$. Where F is the standard normal CDF.
4. Compute marginal inverse quantiles $u_1 \dots u_d = \tilde{F}^{-1}(z_1) \dots \tilde{F}^{-1}(z_d)$. Where $\tilde{F}$ is the empirically estimated CDF of every sampled quantity.

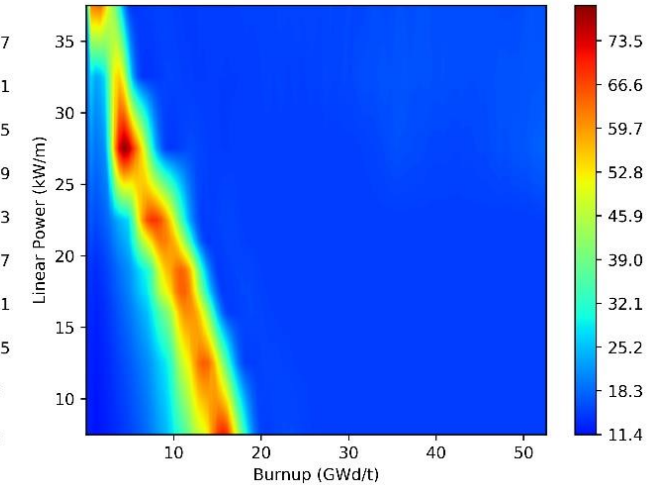
FUEL PERFORMANCE Hi2Lo – RESULTS

- Based on $M = 100$ UTAB samples the RMSE is computed between the predicted H_{gap} by the high fidelity code and the interpolated one from MTAB and UTAB.
- The depletion is performed at mid points of the linear power discretization.
- UTAB performs significantly better than MTAB, reducing the error at least by half in most of the grid points. Especially after the closure the error decreases from $\sim 15\%$ to $\sim 2\%$. This provides also the error associated with each grid point.

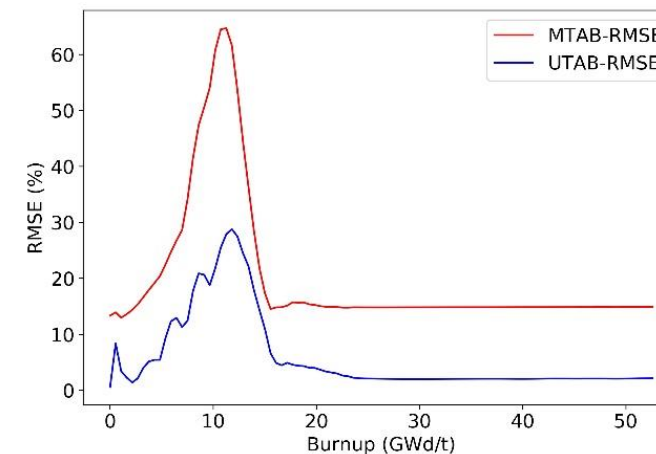
UTAB RMSE in %



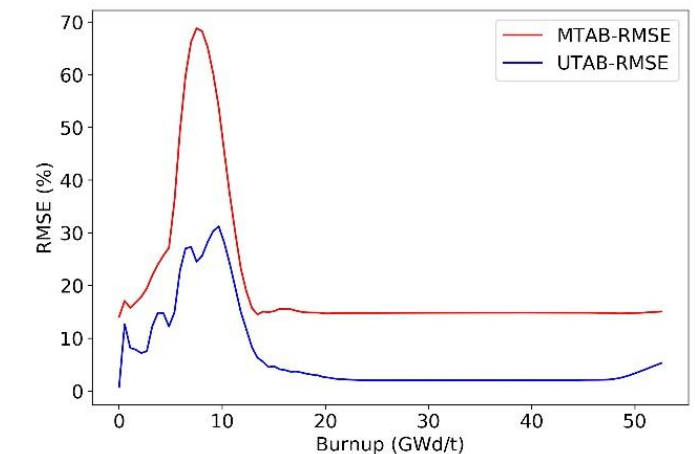
MTAB RMSE in %



RMSE at 17.5kW/m



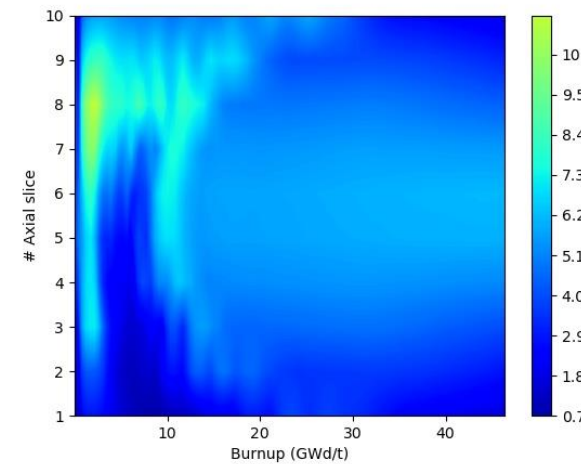
RMSE at 22.5kW/m



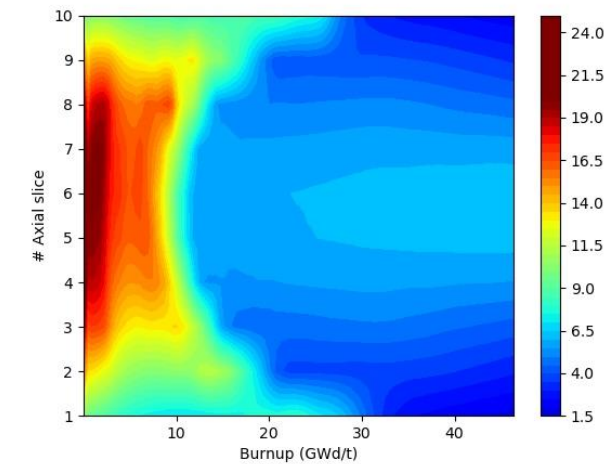
FUEL PERFORMANCE Hi2Lo – RESULTS

- A total of 100 samples are evaluated with CTFFuel using three H_{gap} models: UTAB, MTAB (no uncertainties) and the dynamic model internal to CTFFuel.
- The fuel centerline temperature (FCT) and fuel surface temperature (FST) are evaluated. The results for the $17.5 \frac{\text{kW}}{\text{m}}$ case are shown.
- UTAB outperforms MTAB by $\sim 50\text{K}$ for FCT upper quantile and by 30K for FST.

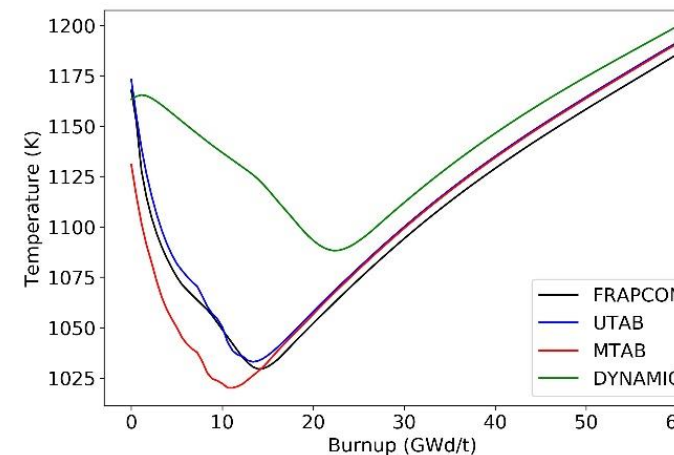
UTAB FCT RMSE in K



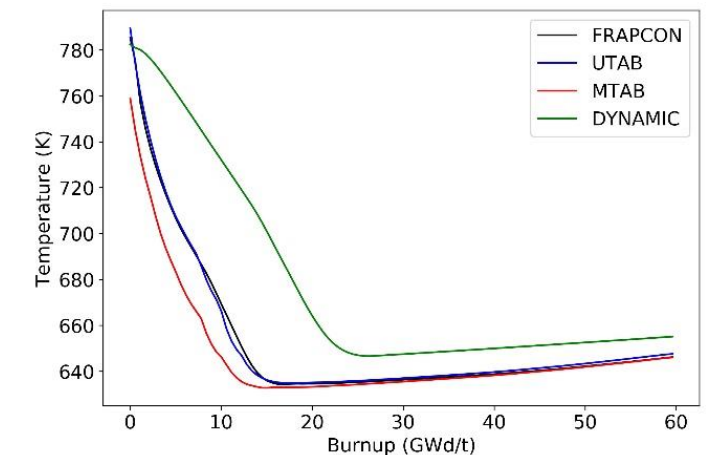
MTAB FCT RMSE in K



FCT upper quantile

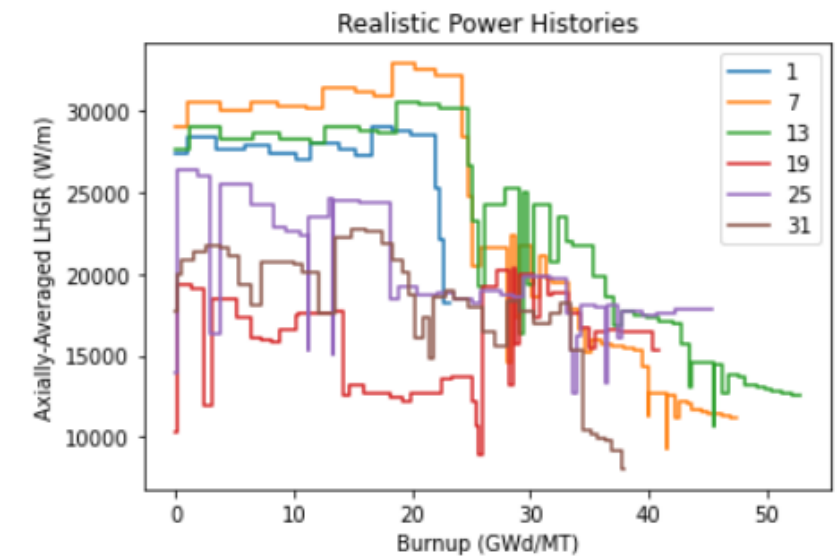


FST upper quantile

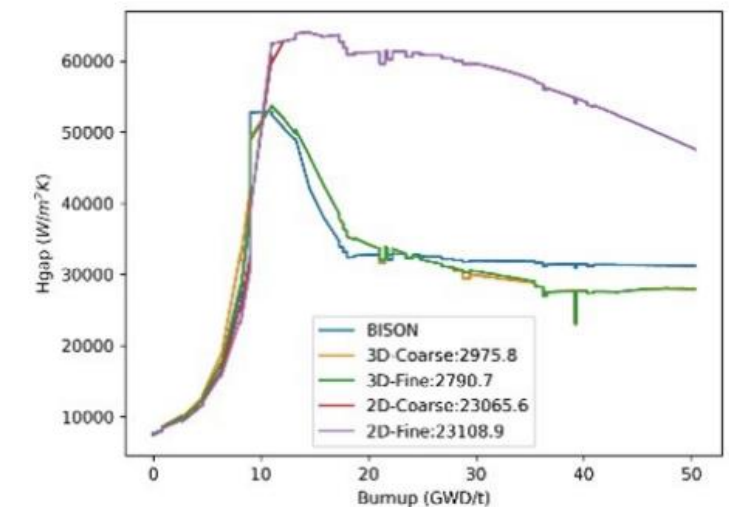


FUEL PERFORMANCE Hi2Lo – UTAB EXTENSION

- Most H_{gap} tables use 2 parameter that account for instantaneous conditions (power) and irradiation conditions (burnup).
- The irradiation history however is usually neglected and could lead to significant discrepancies.
- This study investigates the impact of adding a third parameter in the UTAB tables.
- Realistic power profiles were selected to represent the power history typically seen by a fuel rod.
- The third parameter selected for the studies was the average power up to every depletion step.
- The study compared the 3D/2D tables and the impact of using a fine/coarse discretization mesh.
- The 3D/2D shows a larger impact than fine/coarse.
- Including the third parameter improves significantly power histories with large deviations between cycles (history 7) and has a minimal effect on small deviations (history 25).

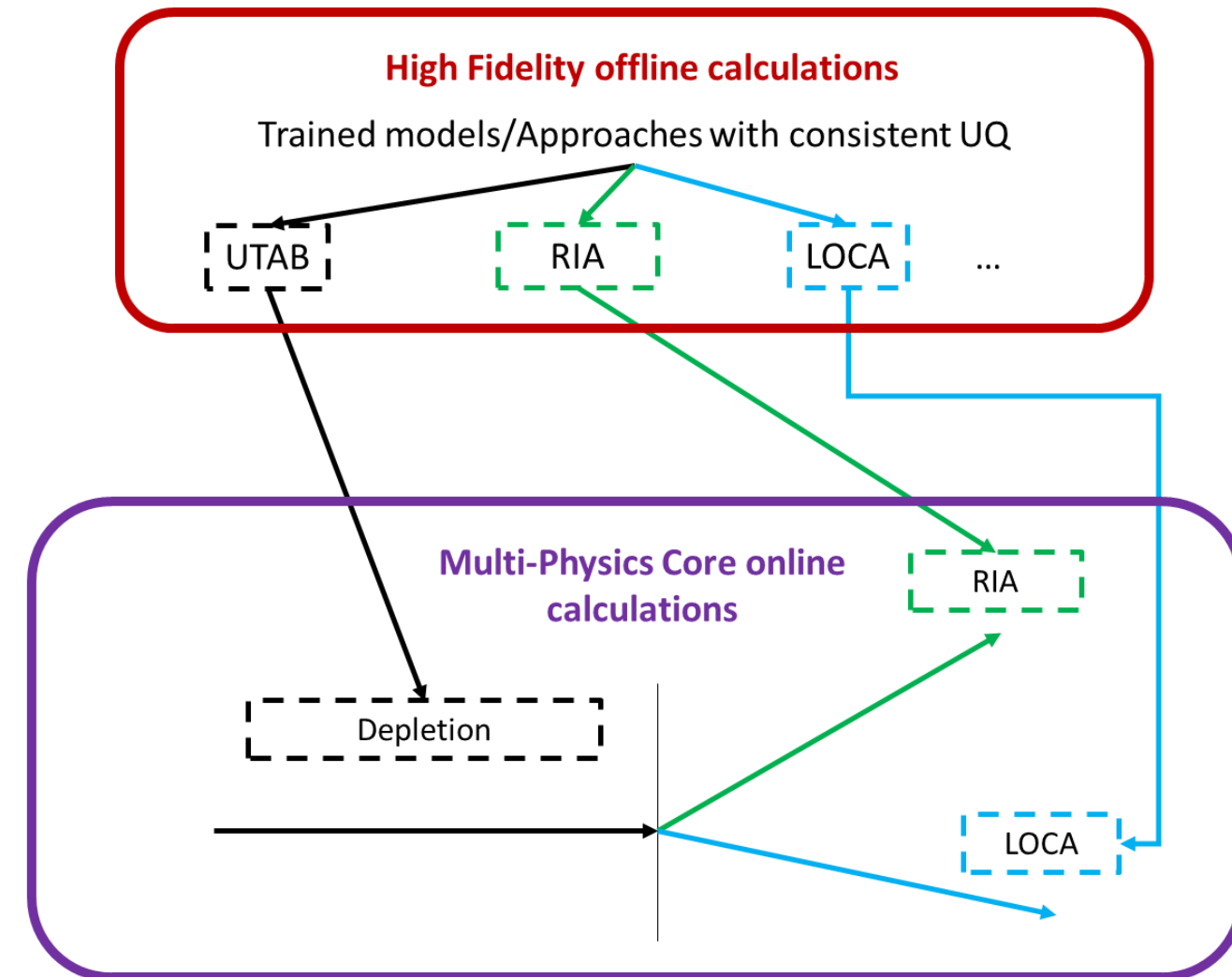


Hgap for history 7



FUEL PERFORMANCE Hi2Lo – SUMMARY AND FUTURE WORK

- Development of an offline Hi2Lo fuel performance approach, called UTAB, that includes a consistent treatment of uncertainties.
- UTAB is an extension of well known tabulated approaches for gap conductance.
- UTAB is applicable to depletion.
- Results show significant improvement in the gap conductance prediction that translates to a moderate improvement in temperature predictions.
- UTAB was implemented in CTFFuel including the capability for a third parameter.
- Future work will focus on developing Hi2Lo gap conductance models for transients.

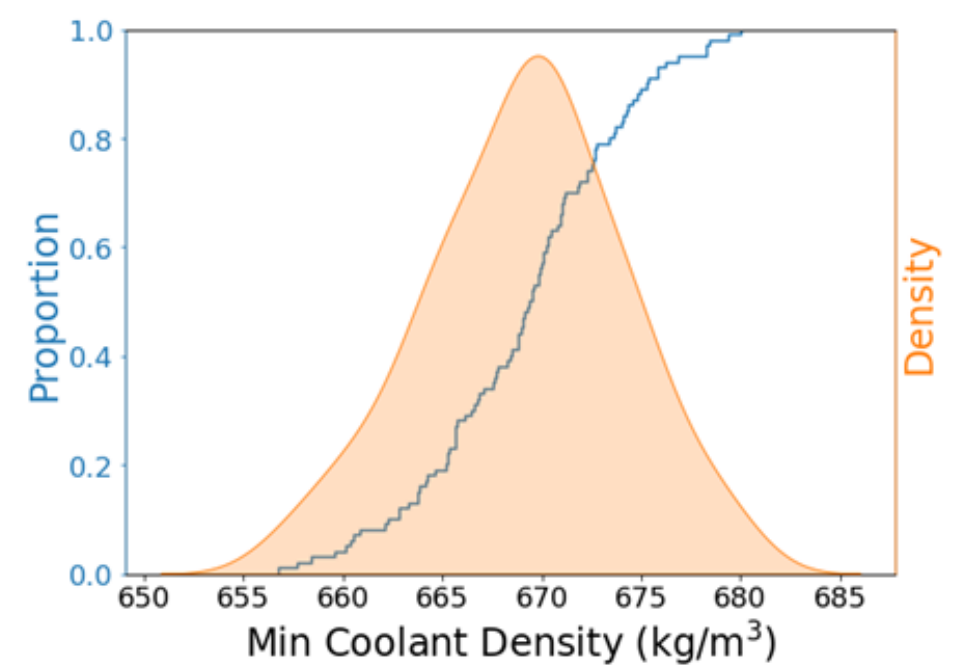
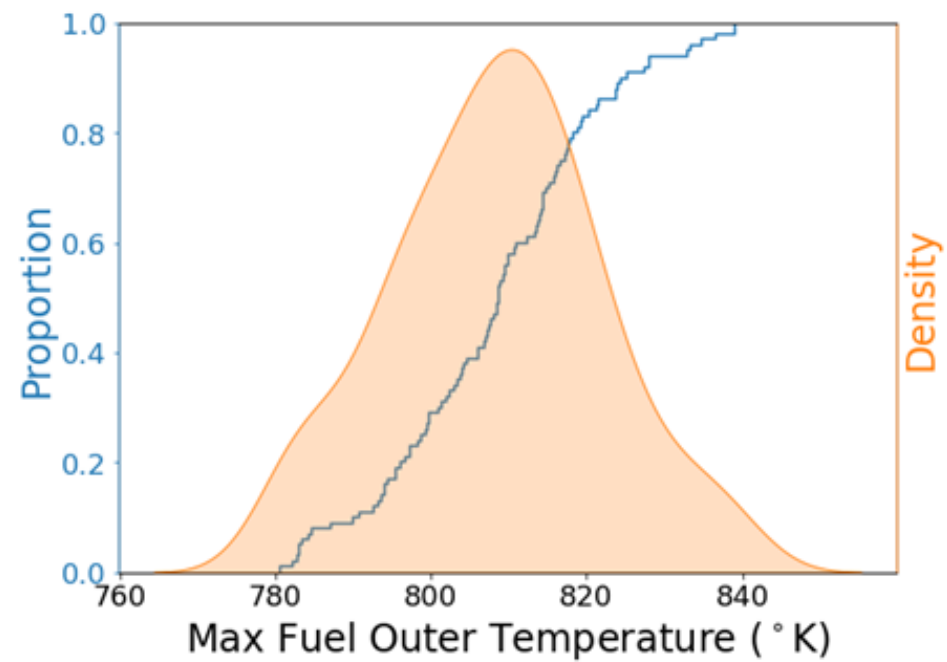
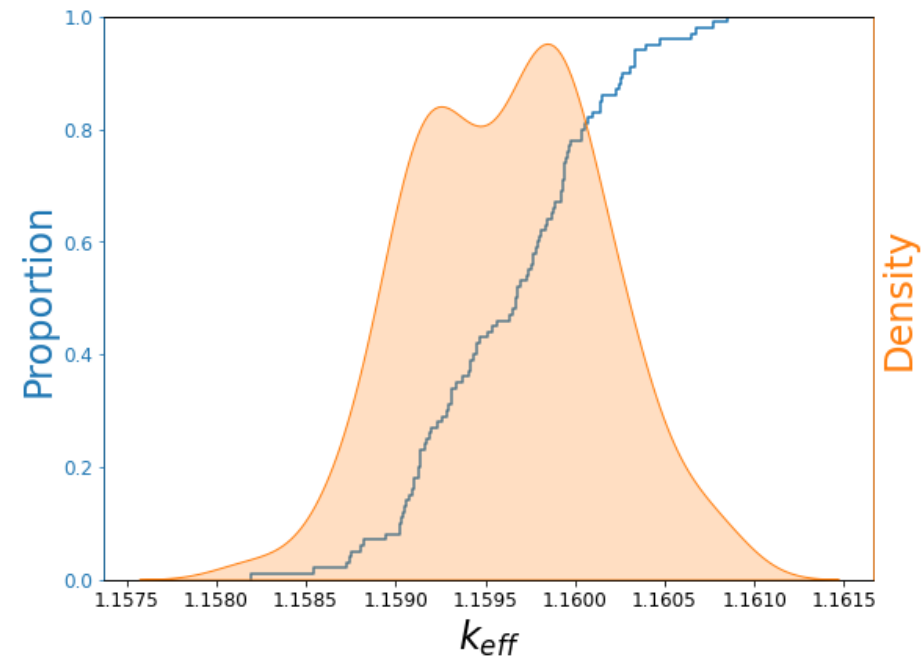


UNCERTAINTY ANALYSIS

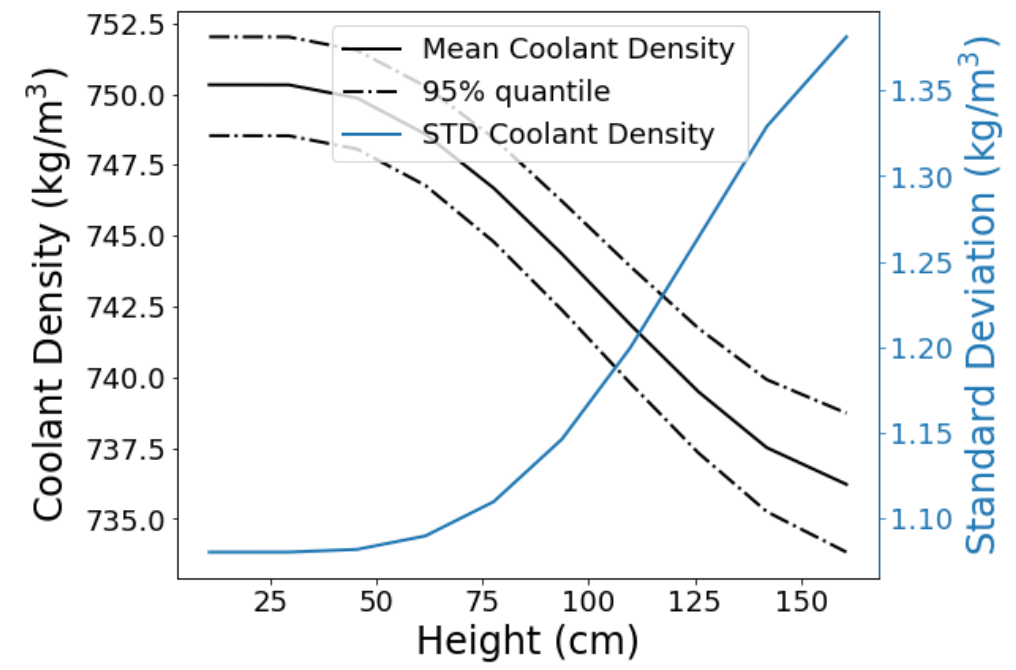
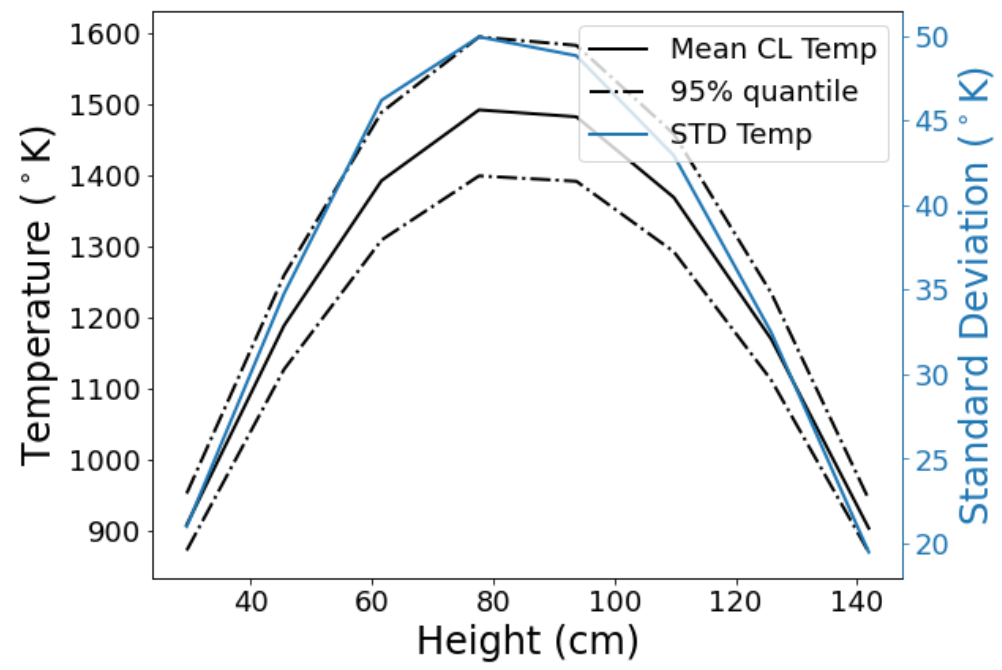
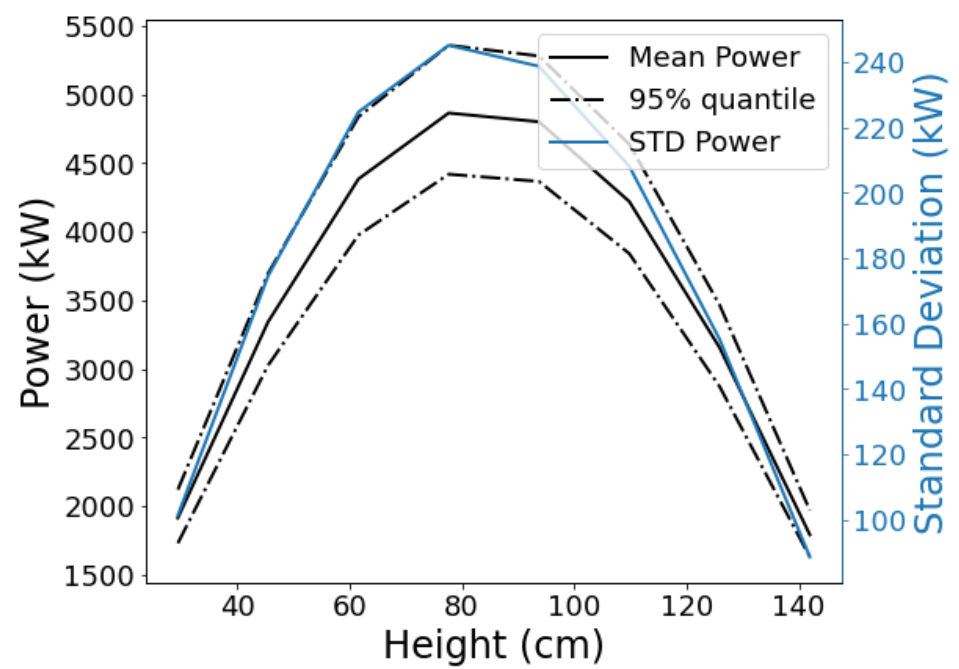
- The uncertainty analysis is performed using 100 random samples drawn for each input of interest from its associated distribution.

Common Inputs		Fuel Performance	
Inlet Temperature (K)	U(560, 562)	Gap Width (μm)	N(85, 7.64)
Inlet Mass Flux (kg/m ² s)	N(3460, 17.3)	Fuel Roughness (μm)	N(2, 0.25)
Total Power (MW)	N(30, 1.5)	Grain size (μm)	N(20, 6)
Inlet Pressure (bar)	N(155.1, 0.517)	Cladding Creep Multiplier	N(1.0, 0.14)
Thermal-Hydraulics		Fission Gas Release Multiplier	LN(1.0,1.0)
Friction B (p=5)	N(18.797, 0.941)		

UNCERTAINTY ANALYSIS

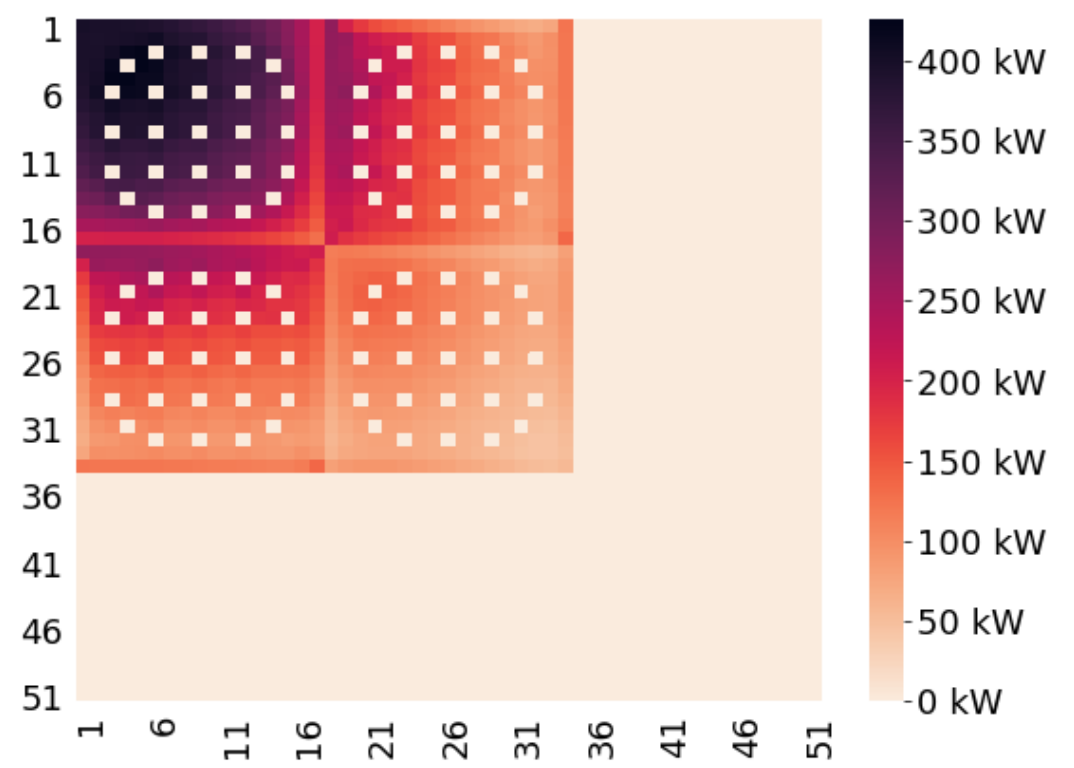
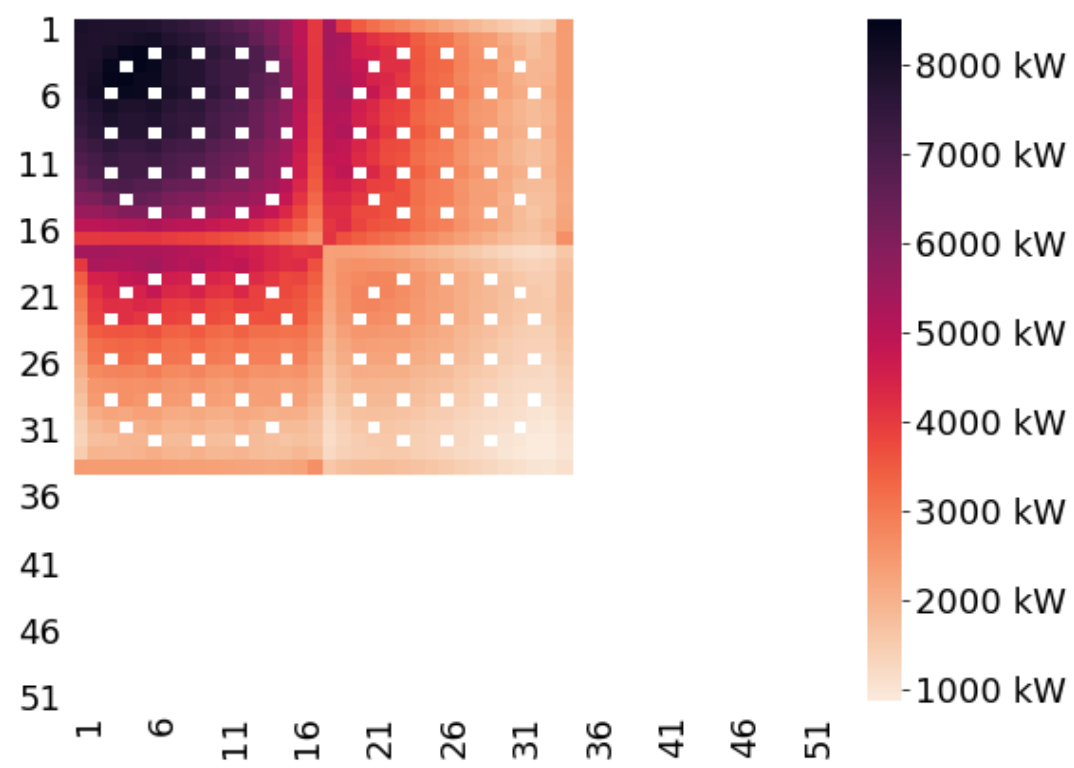


UNCERTAINTY ANALYSIS



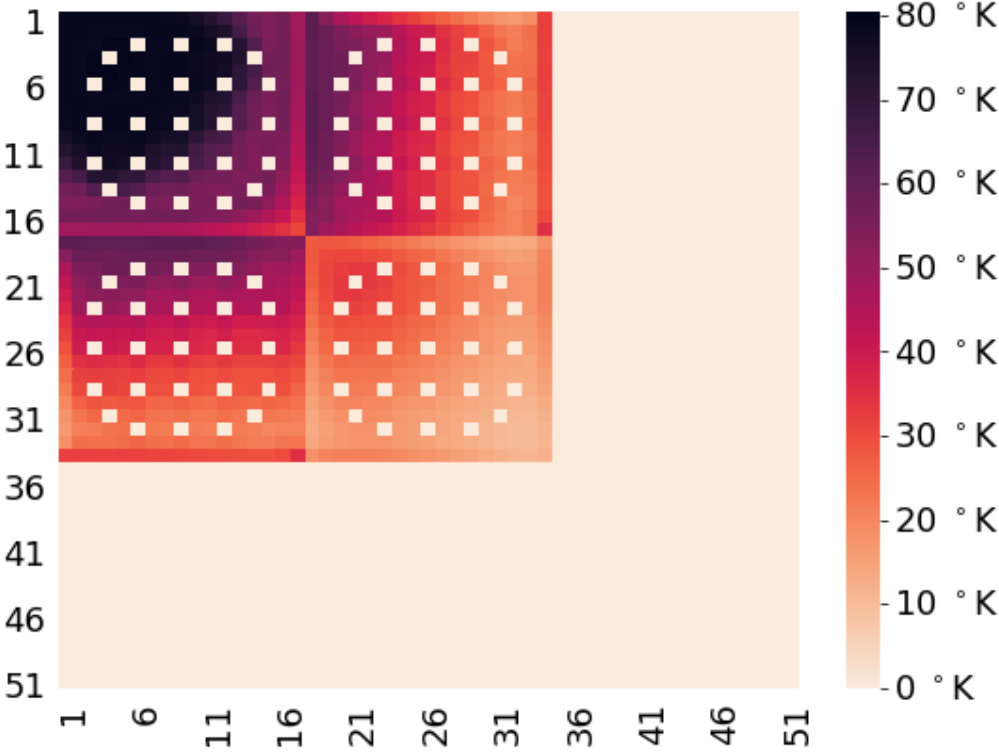
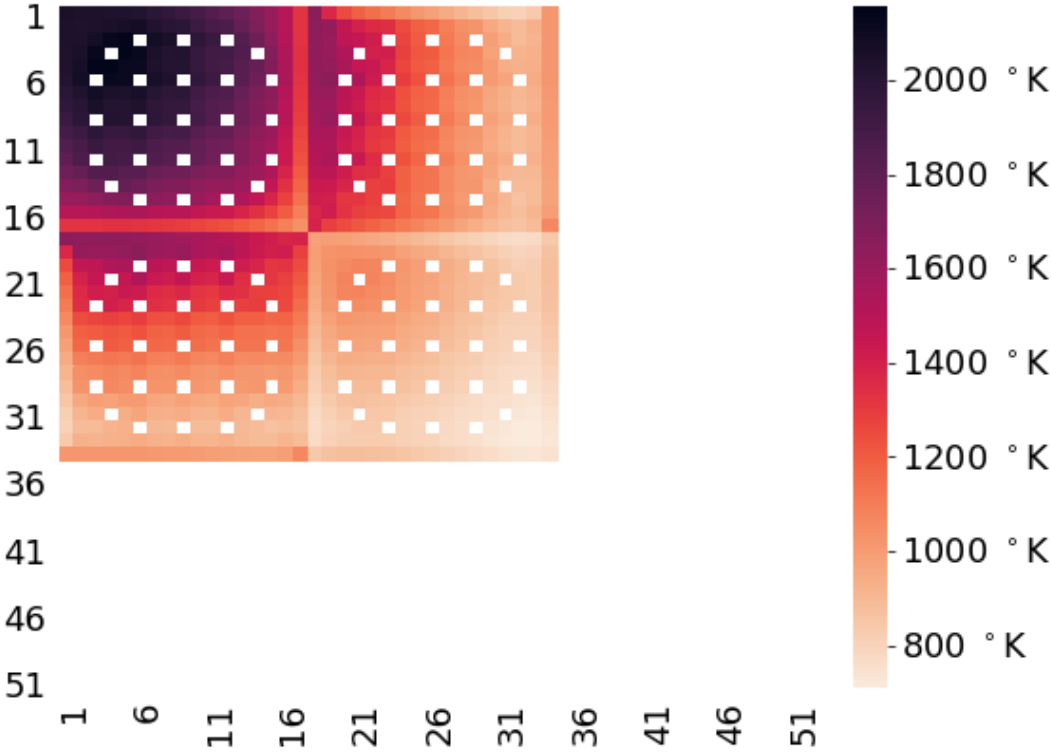
UNCERTAINTY ANALYSIS

Radial power mean (left) and uncertainty in the form of standard deviation (right)



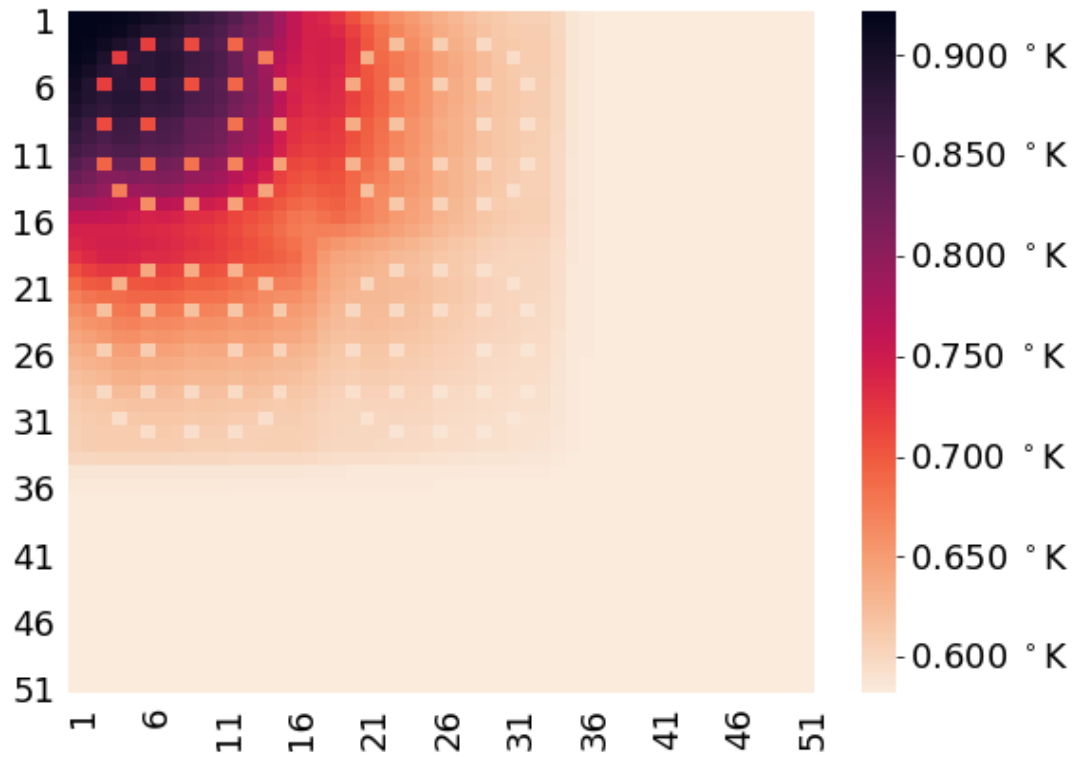
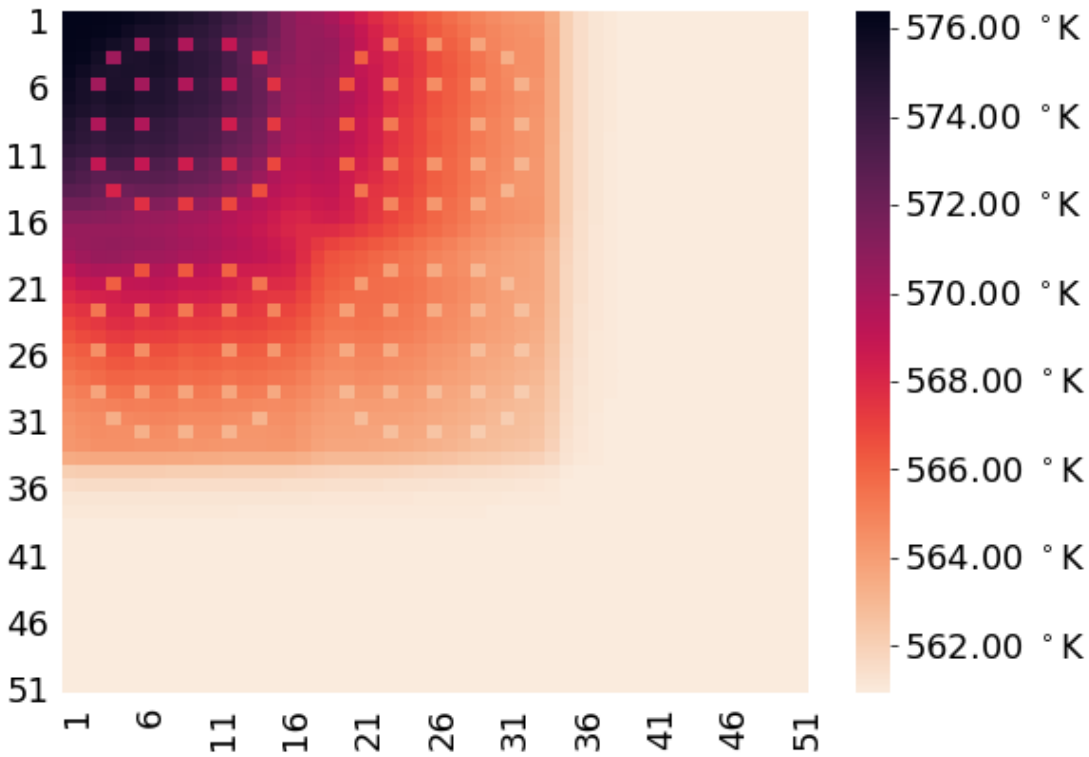
UNCERTAINTY ANALYSIS

Fuel centerline temperature mean (left) and uncertainty in the form of standard deviation (right)



UNCERTAINTY ANALYSIS

Coolant temperature mean (left) and uncertainty in the form of standard deviation (right)



Next Class: *Spatial & Temporal Coupling - Multi-Physics Platforms*